At the Break of Cosmic Dawn: Identifying and Understanding the Most Distant Galaxies with JWST

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PennState Eberly College of Science

JADES JWST ADVANCED DEEP EXTRAGALACTIC SURVEY

What are galaxies?

What is the James Webb Space Telescope (JWST)?

How is JWST being used to study the most distant galaxies?

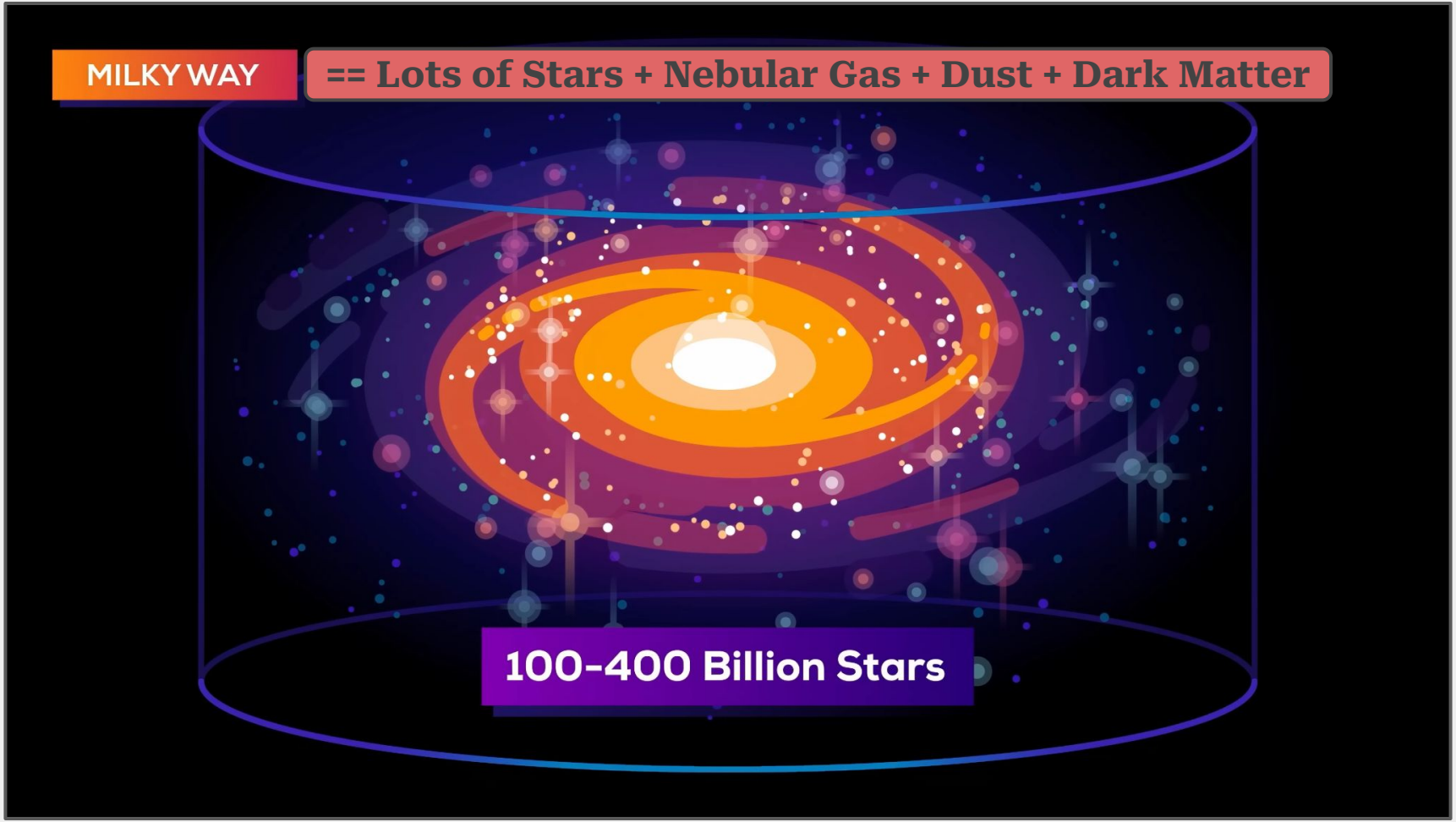
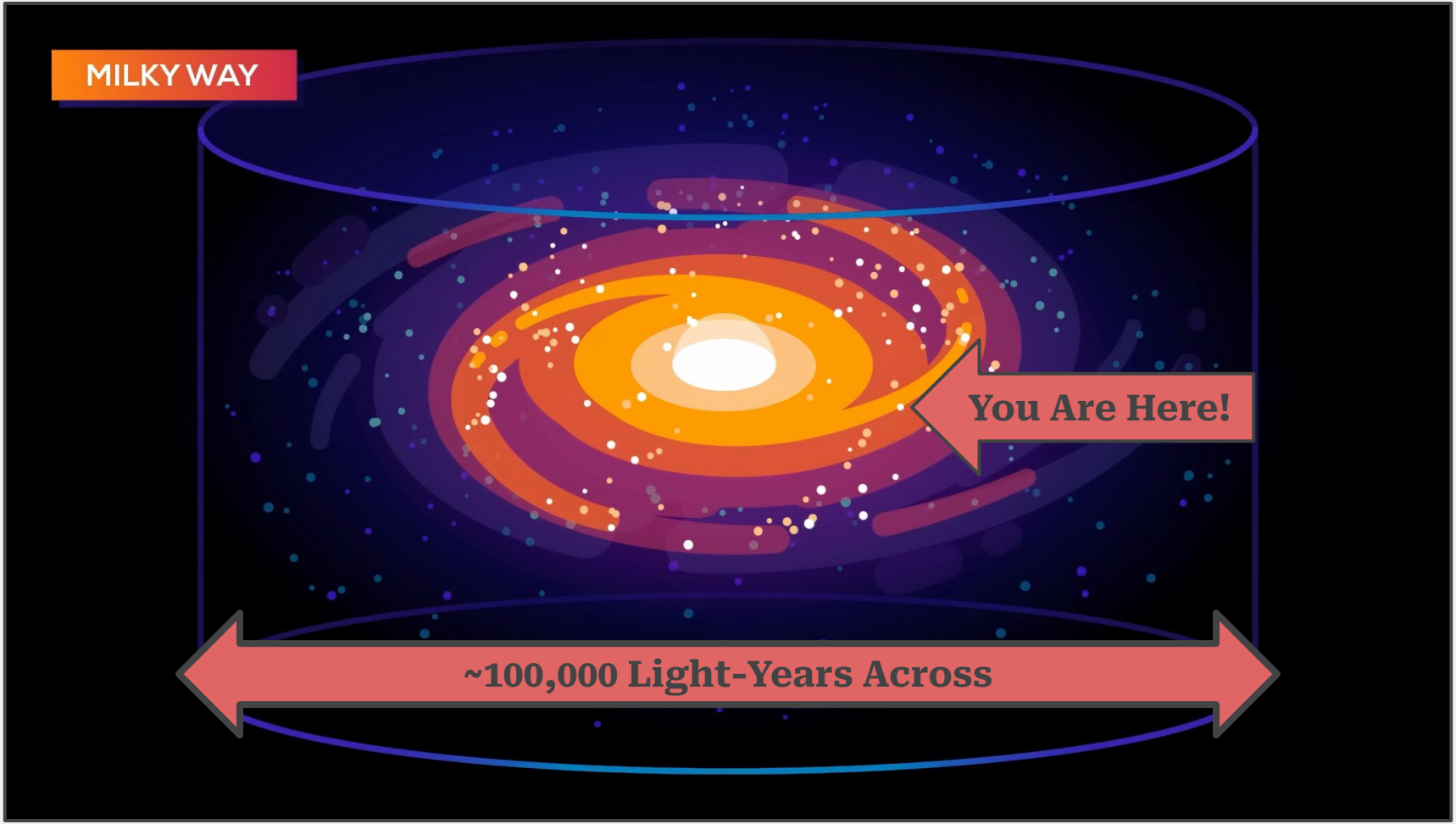


Image Credit: [Kurzgesagt – In a Nutshell]



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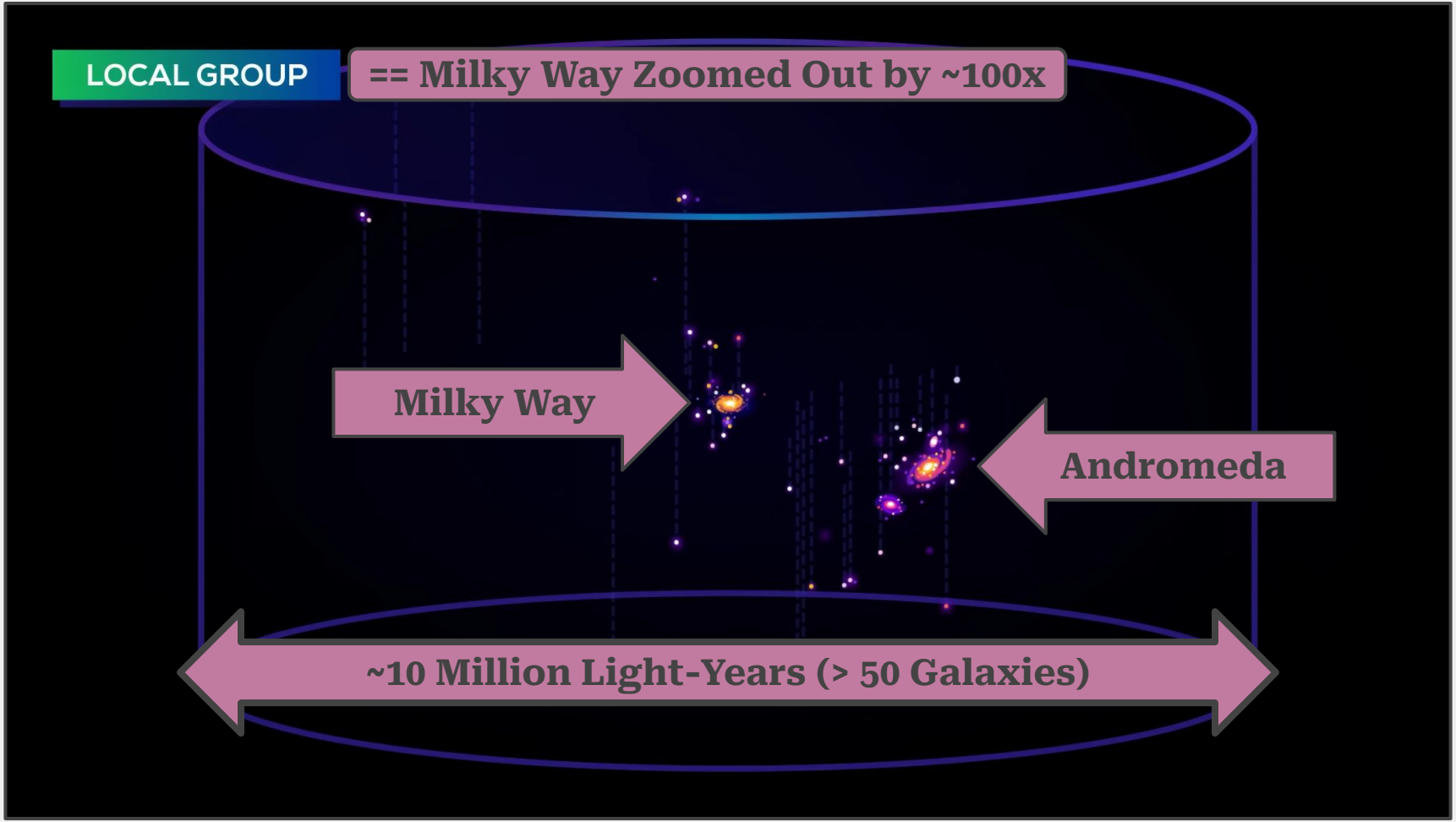

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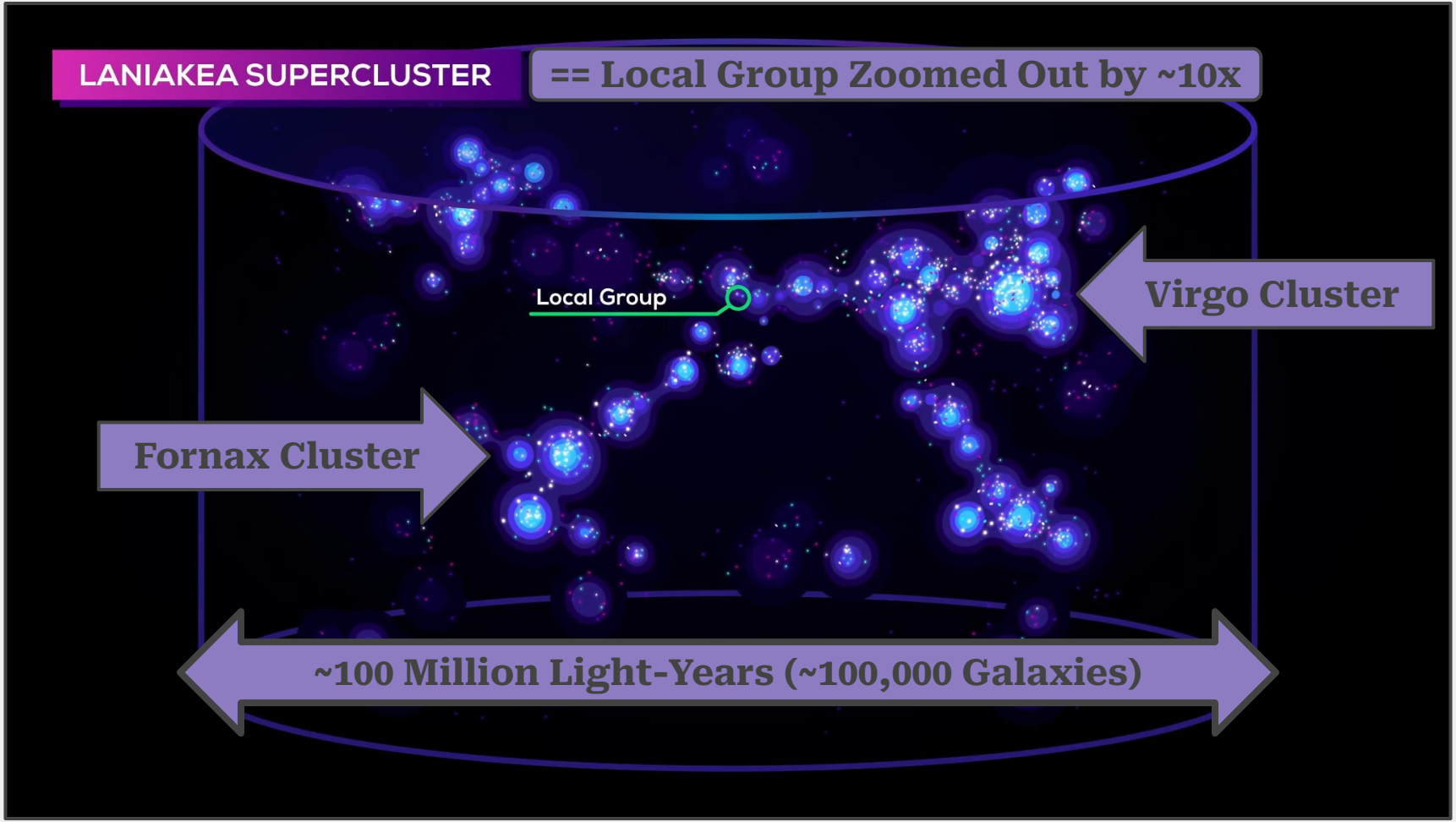

Image Credit: [Kurzgesagt – In a Nutshell]

The "Cosmic Web" (Geller & Huchra 1989)

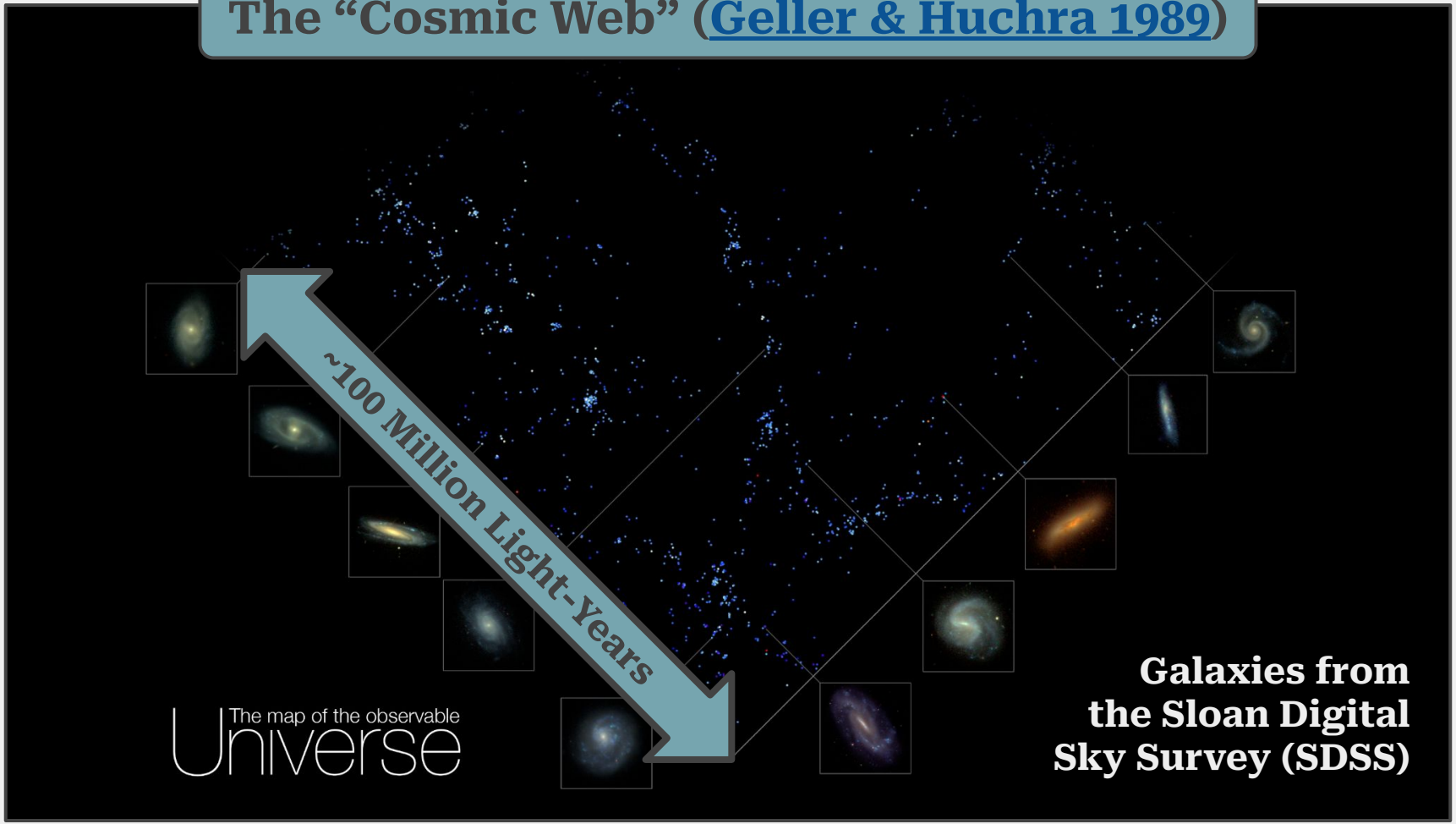

Image Credit: B. Ménard and N. Shtarkman (Johns Hopkins University)

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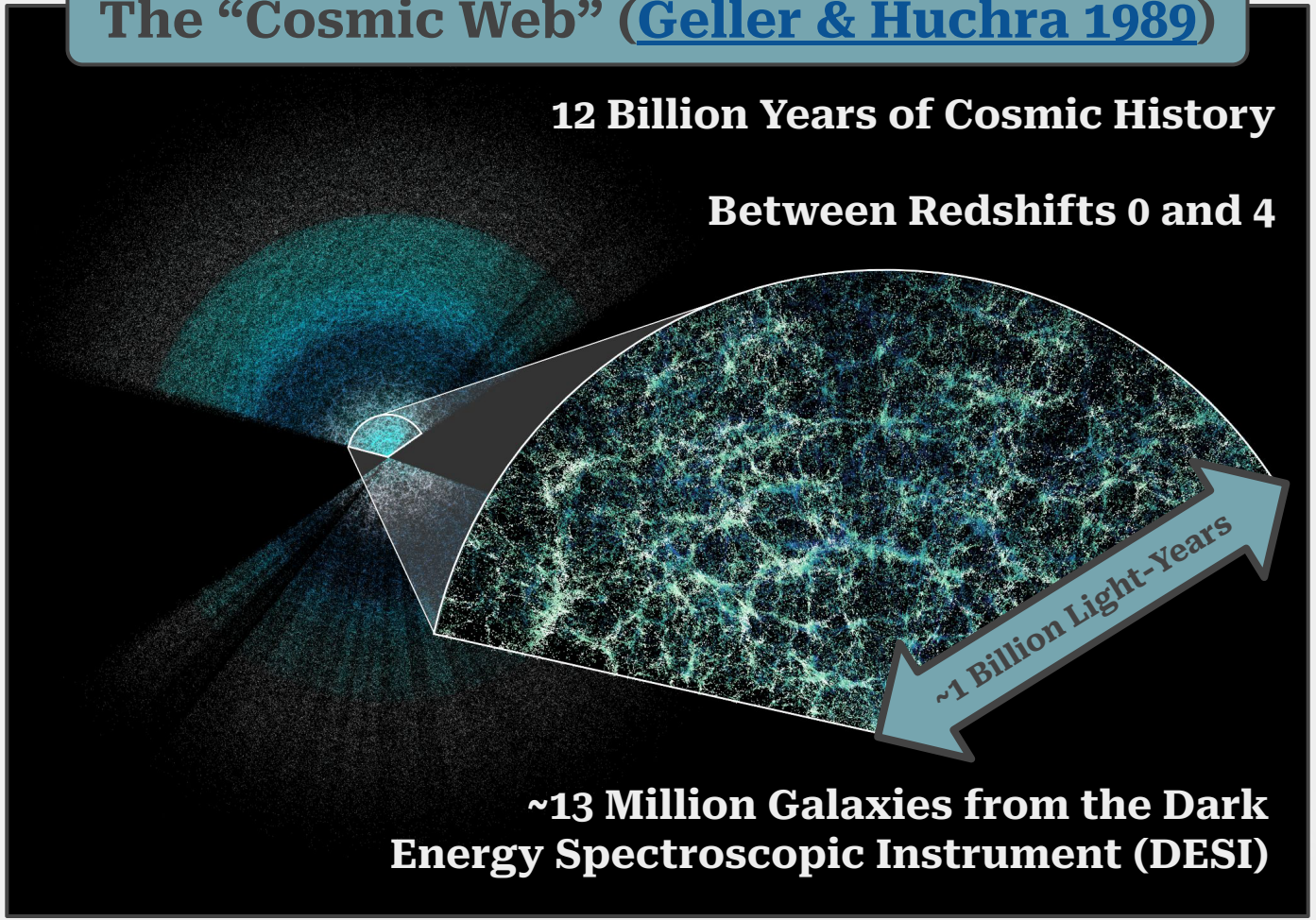


Image Credit: C. Lamman (Harvard University) and the DESI Collaboration

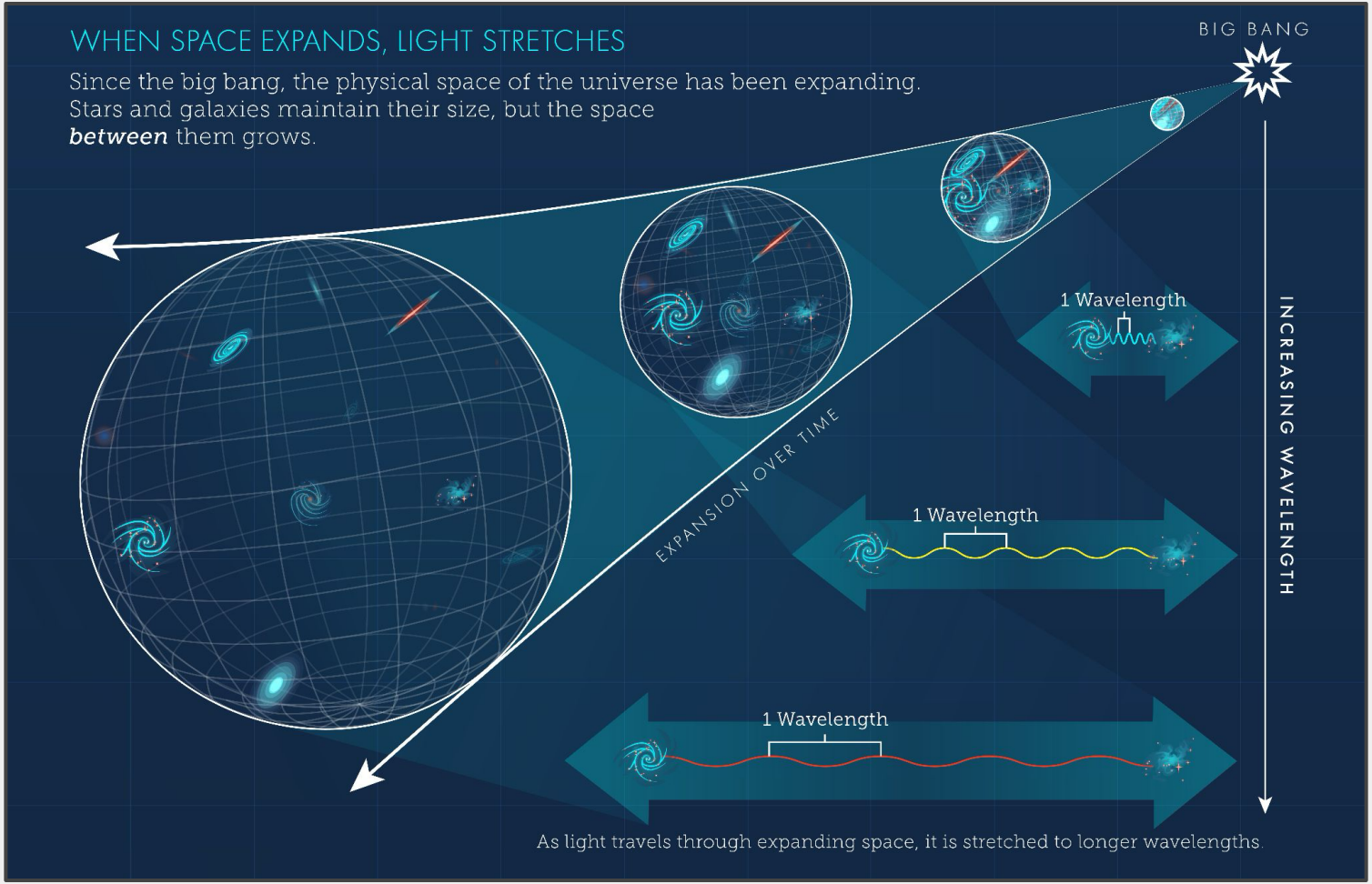
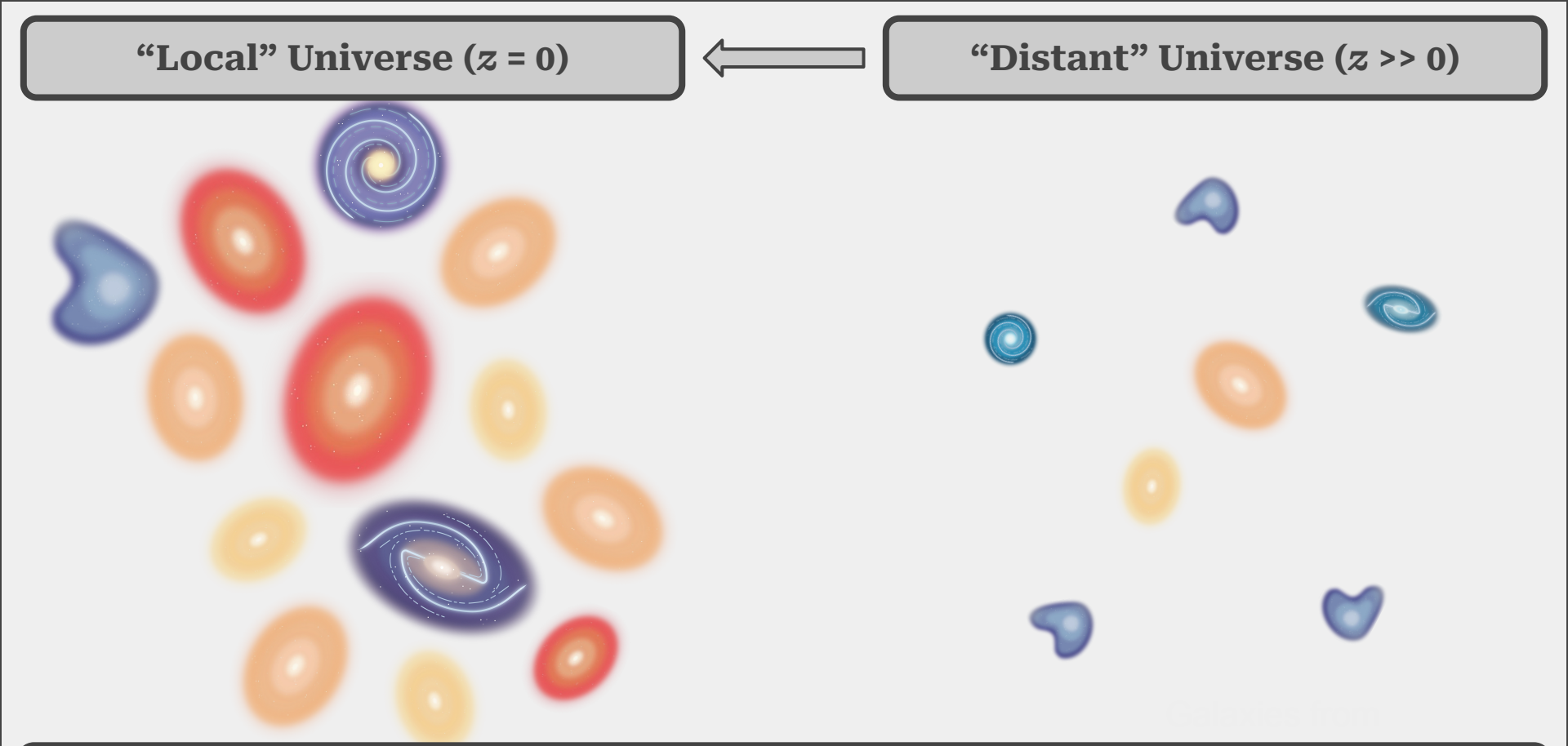


Image Credit: NASA, ESA, L. Hustak (STScI)

“Local” Universe ($z = 0$)

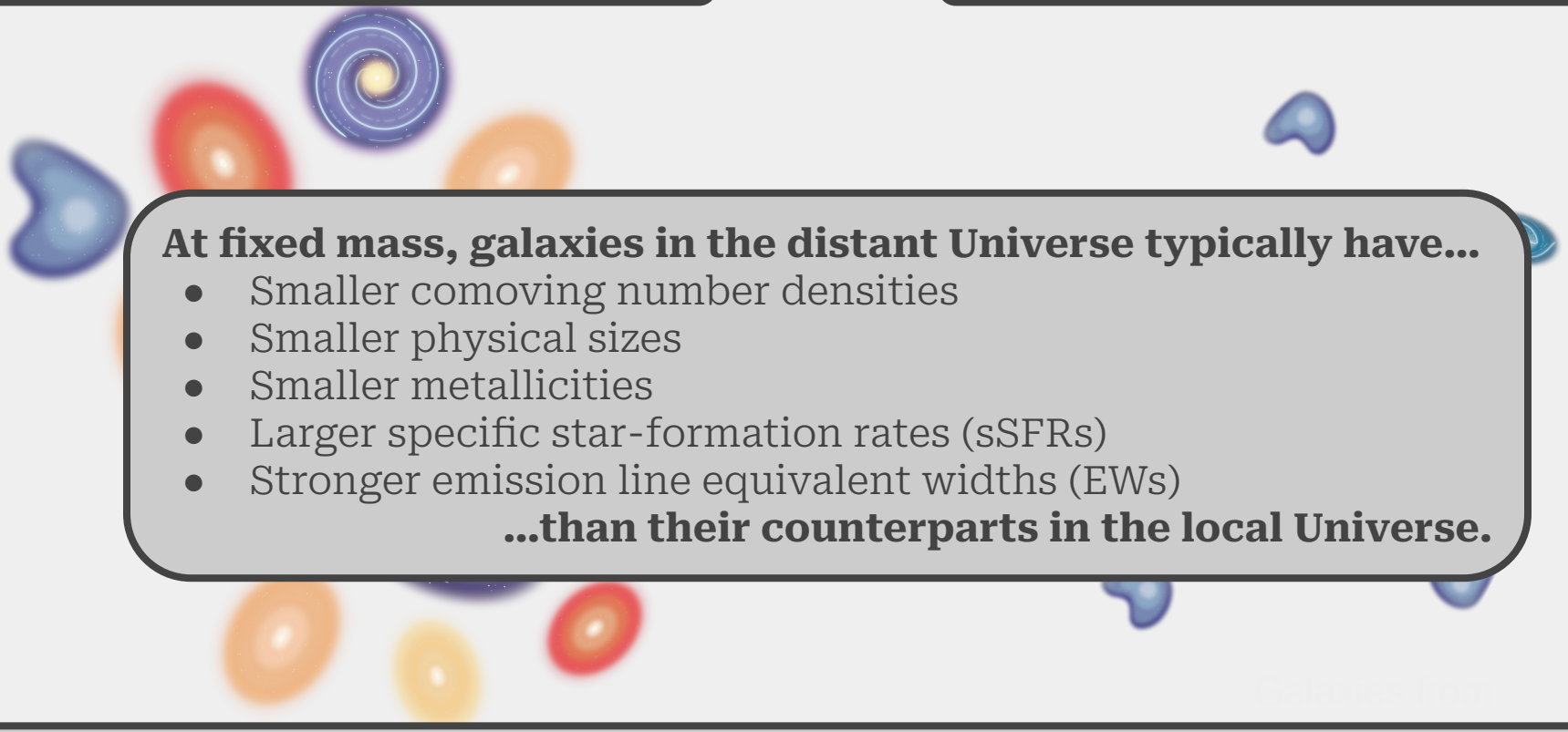


“Distant” Universe ($z >> 0$)

Distant galaxies provide powerful constraints on galaxy formation theories!

Image Credit: NASA

“Local” Universe ($z = 0$) ⟸ **“Distant” Universe ($z >> 0$)**



At fixed mass, galaxies in the distant Universe typically have…

- Smaller comoving number densities
- Smaller physical sizes
- Smaller metallicities
- Larger specific star-formation rates (sSFRs)
- Stronger emission line equivalent widths (EWs)

…than their counterparts in the local Universe.

Distant galaxies provide powerful constraints on galaxy formation theories!

Image Credit: NASA

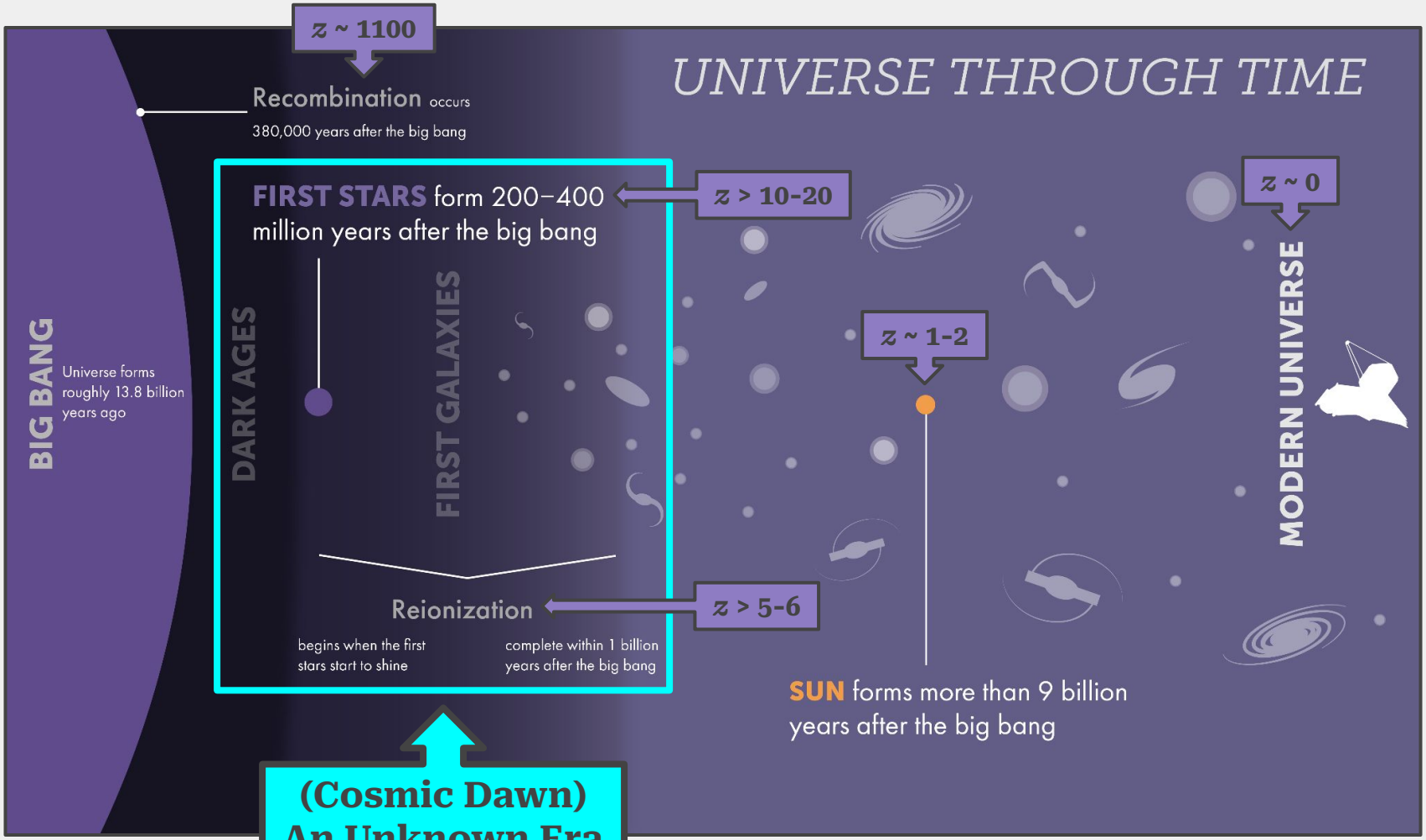


Image Credit: NASA, ESA, CSA, STScI

~~What are galaxies?~~

What is the James Webb Space Telescope (JWST)?

How is JWST being used to study the most distant galaxies?

Webb == JWST

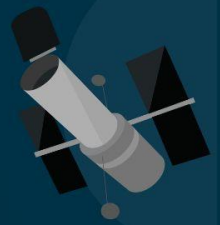
COMPARING WEBB AND HUBBLE

Webb follows the NASA/ESA Hubble Space Telescope in the line of great space observatories. Both space observatories have different capabilities and will operate in parallel, complementing each other.

HUBBLE

570 km

Primary mirror
2.4 m



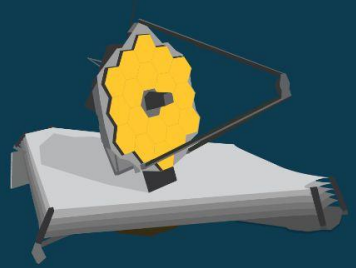
12.5 billion years ago

wavelengths

WEBB

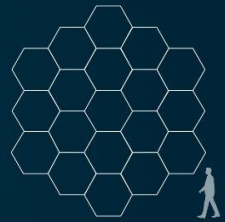
1.5 million km

Second Sun-Earth Lagrange point



Primary mirror
6.5 m

18
mirror segments



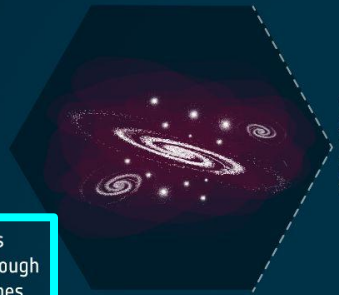
13.5 billion years ago

wavelengths

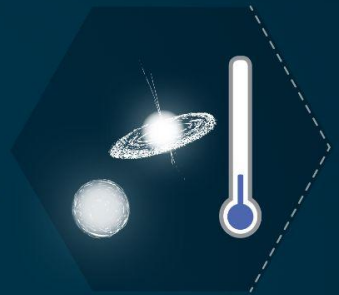
JWST could detect the heat from a bumblebee on the moon!

Image Credit: ESA

WHY STUDY THE UNIVERSE IN INFRARED?



Near-infrared light reveals the formation of galaxies, and peers through the dust layers that enshroud new born stars



Mid-infrared light peers through the cold dusty regions where stars form, and reveals how massive stars and black holes shape their surroundings

The light of galaxies that are billions of light-years away travels to us through space that is expanding. This stretches the light's wavelength into the infrared allowing Webb to see far back in time

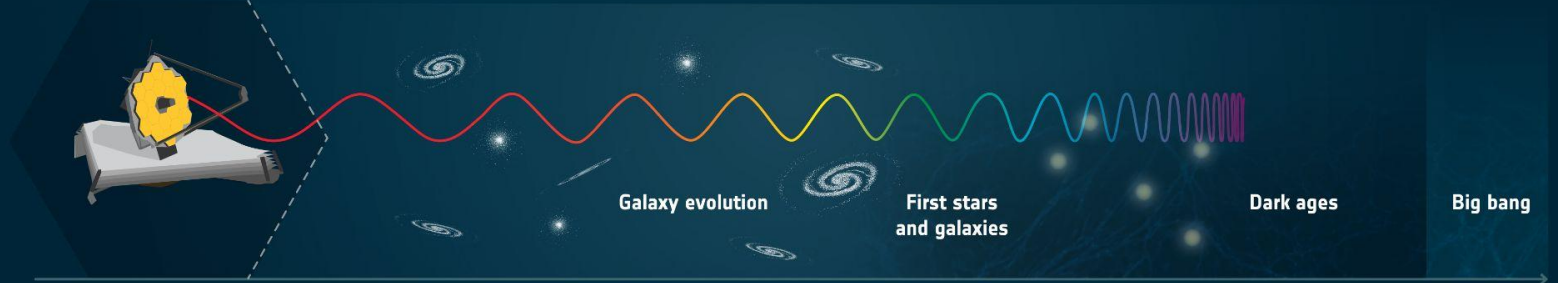


Image Credit: ESA

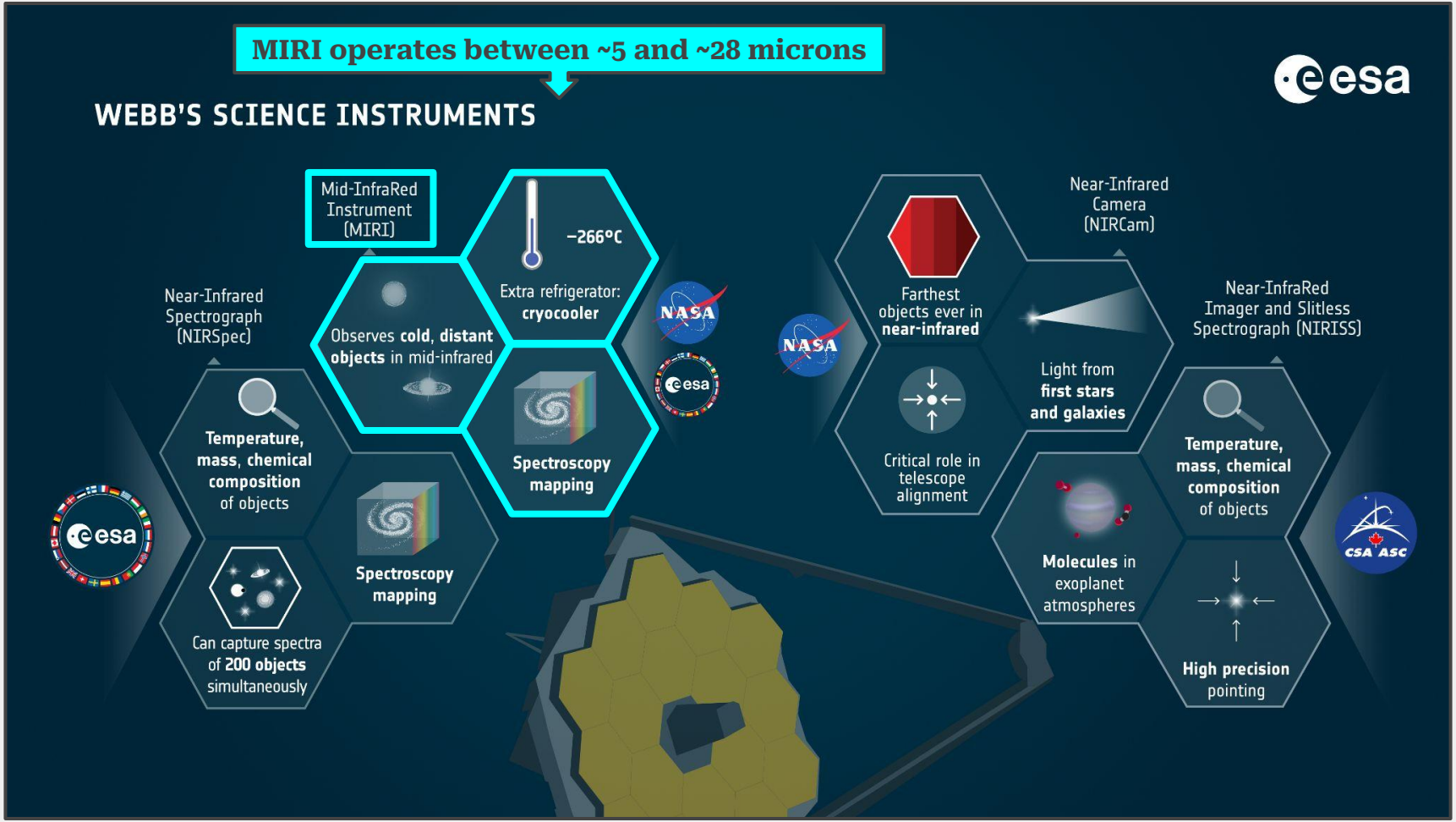


Image Credit: ESA

WEBB SCIENCE

Webb is designed to answer outstanding questions about the Universe and to make breakthrough discoveries in all fields of astronomy.

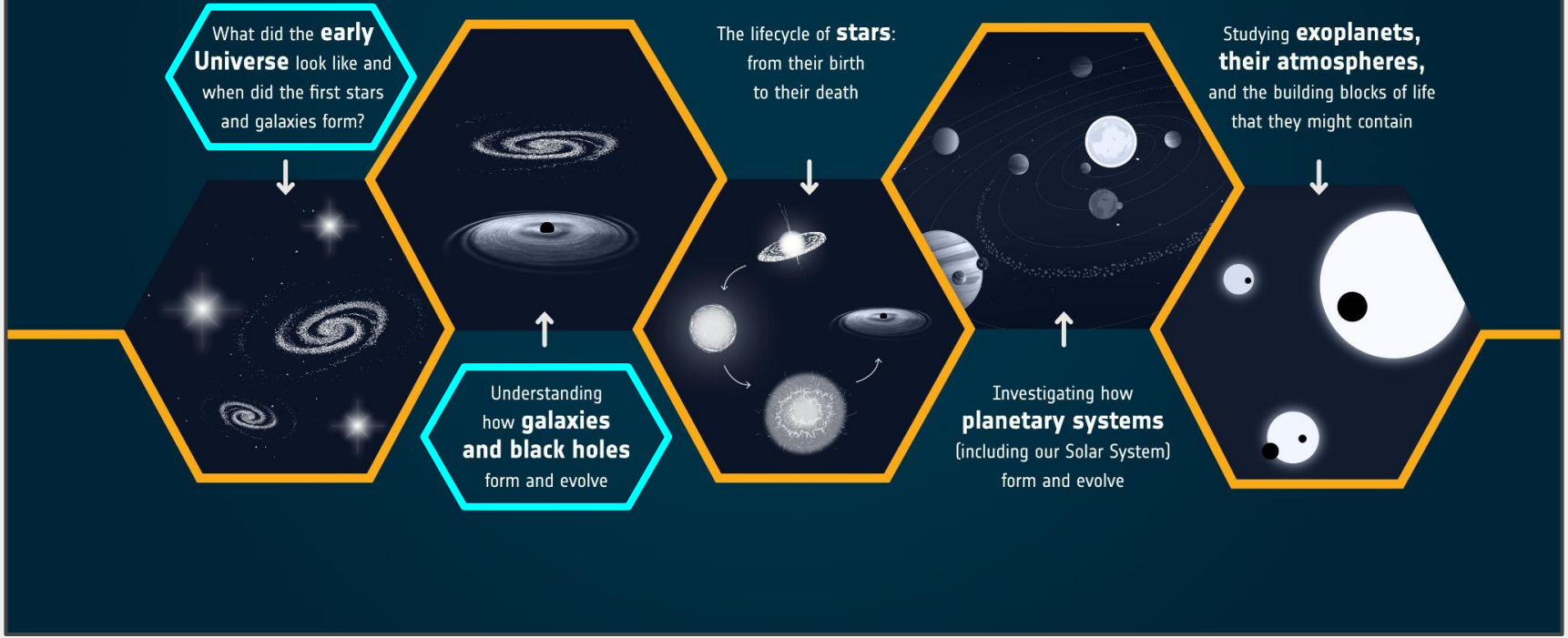
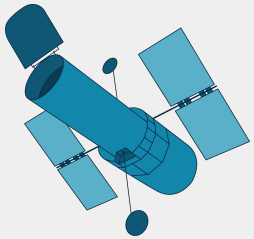
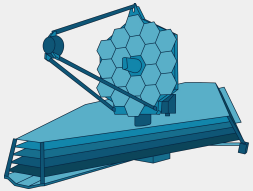


Image Credit: ESA

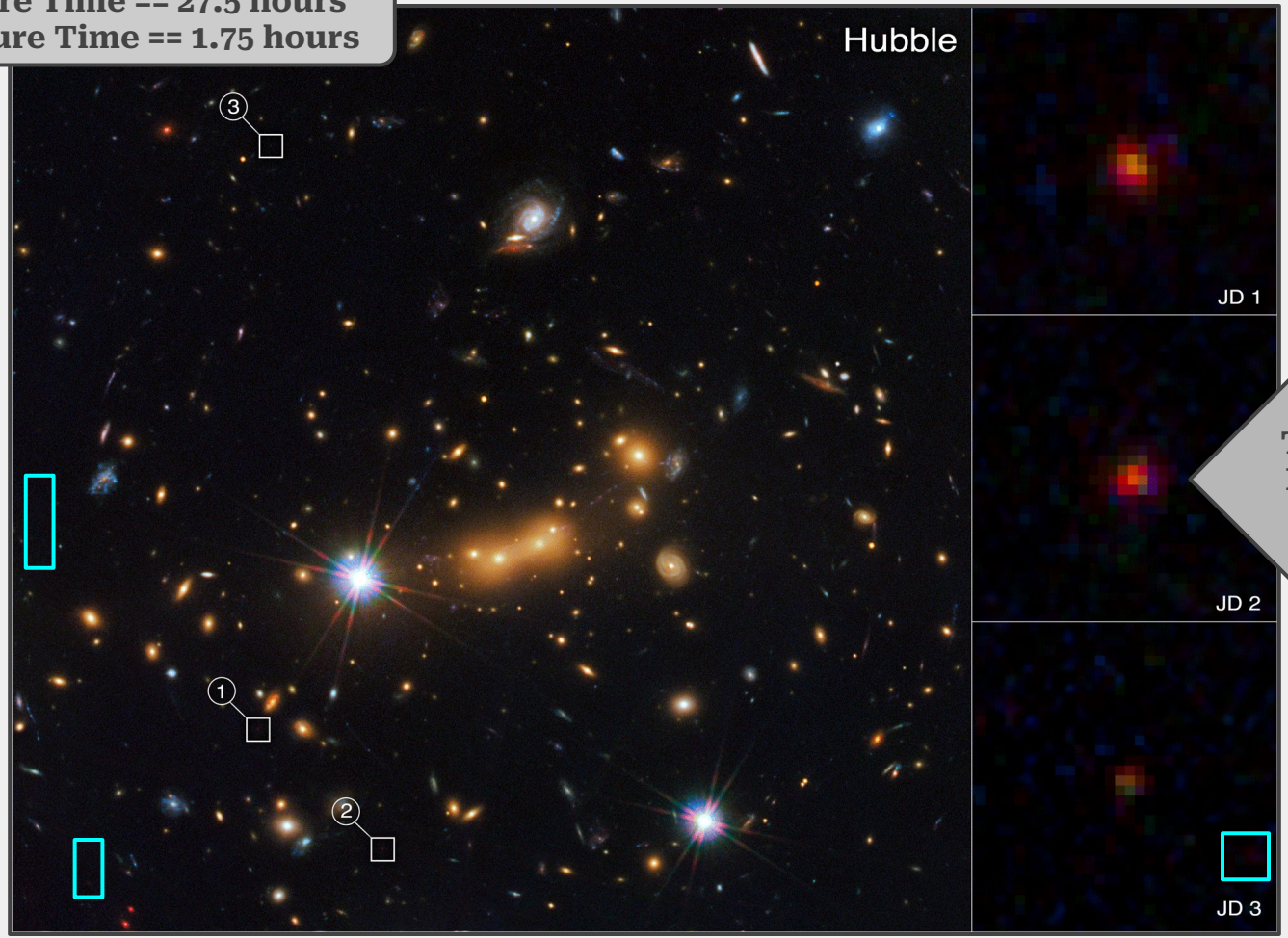
HST Exposure Time == 27.5 hours
JWST Exposure Time == 1.75 hours



HST



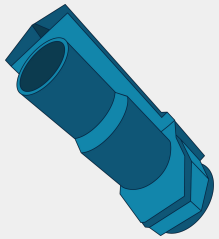
JWST



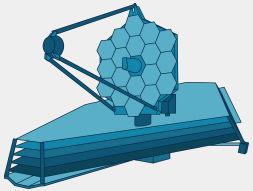
Triply Imaged
MACS0647-JD
z = 10.17

Visualization Credit: NASA, ESA, CSA, A. Pagan (STScI)

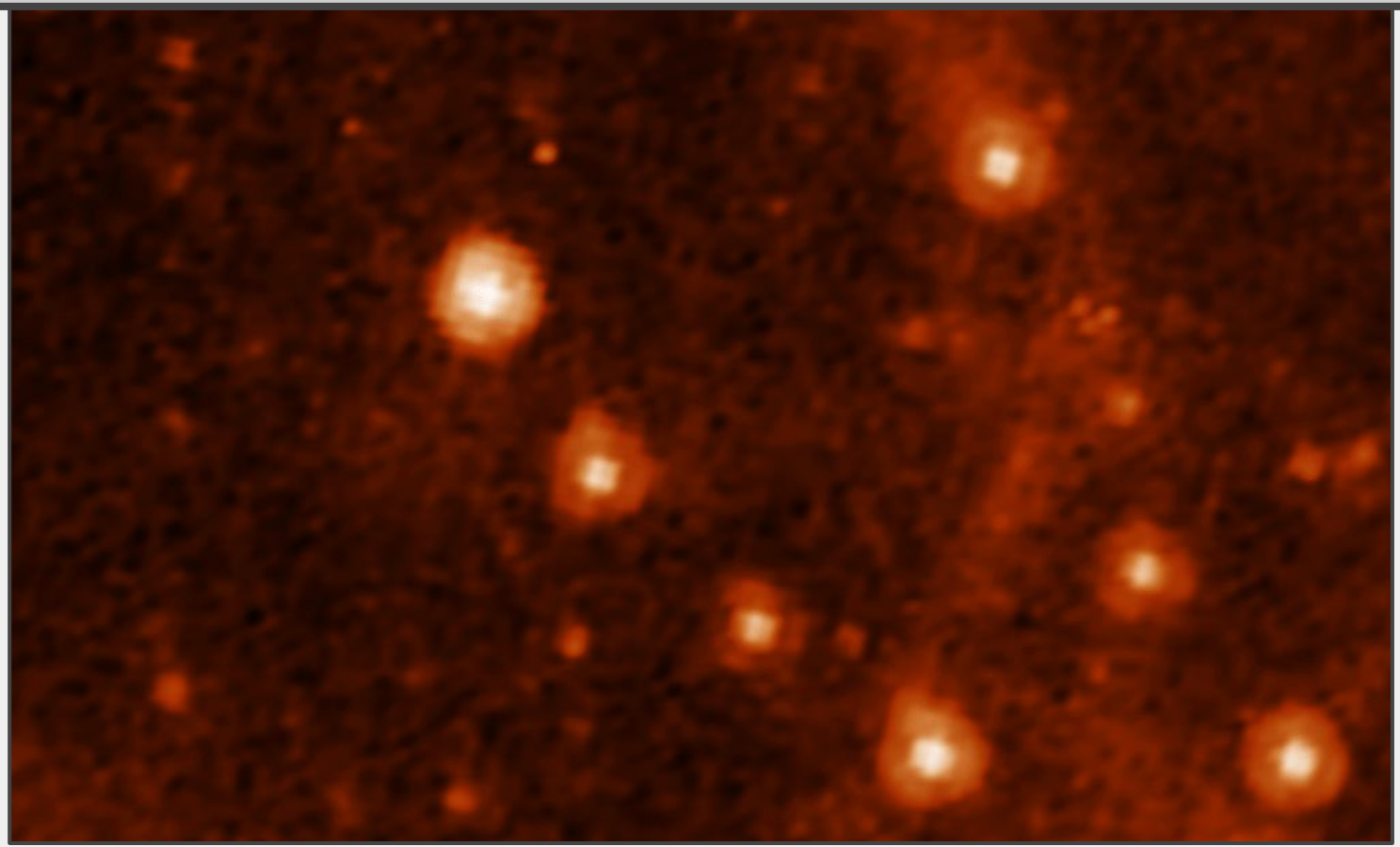
Spitzer/IRAC versus JWST/MIRI at 7.7 microns of the LMC



Spitzer



JWST



Visualization Credit: NASA, JPL/Caltech (Spitzer); NASA, ESA, CSA, STScI (MIRI)

Image Credit: [NASA, ESA, CSA, Northrop Grumman]

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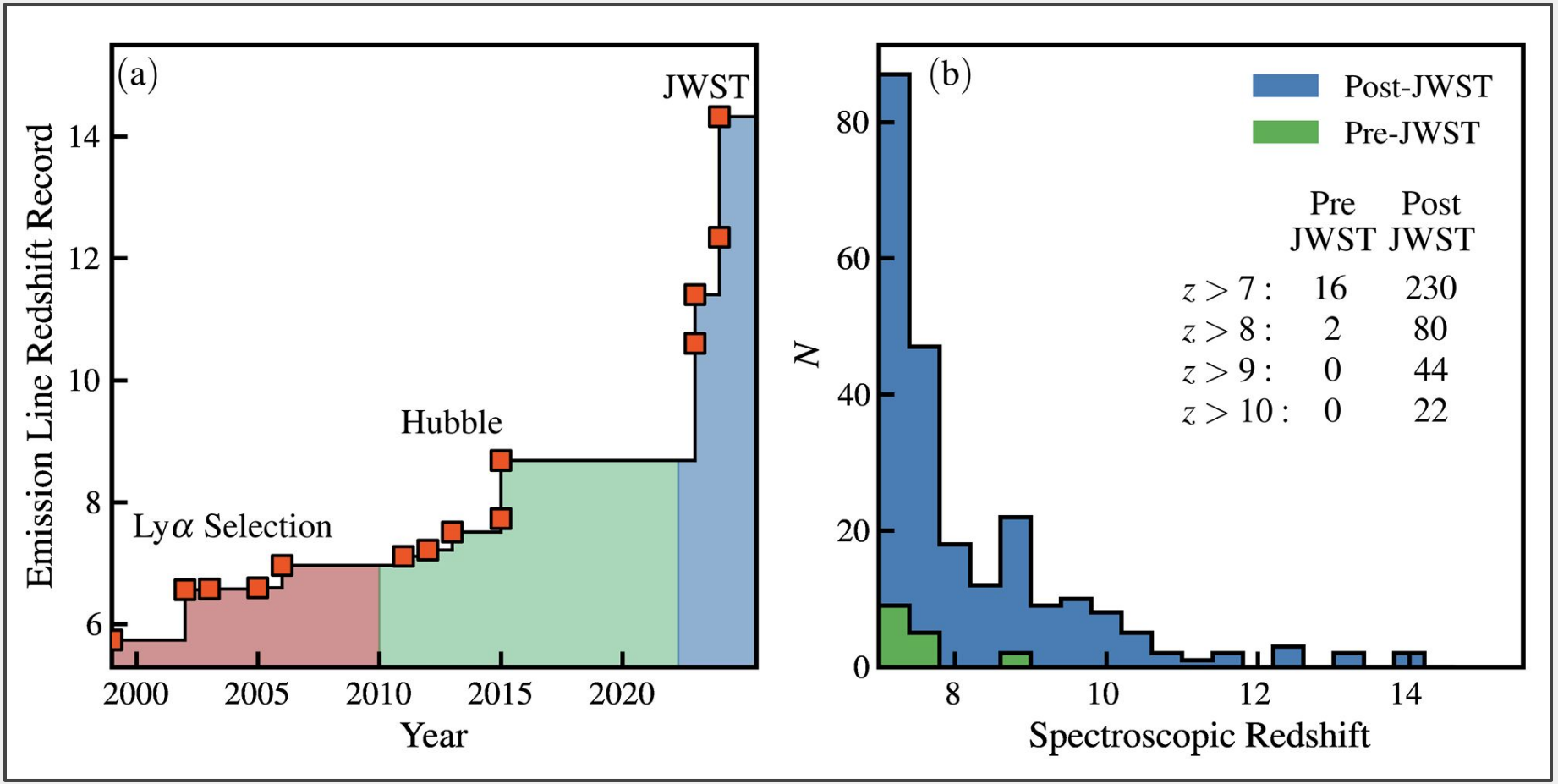


Image Credit: Daniel P. Stark et al. (2026)

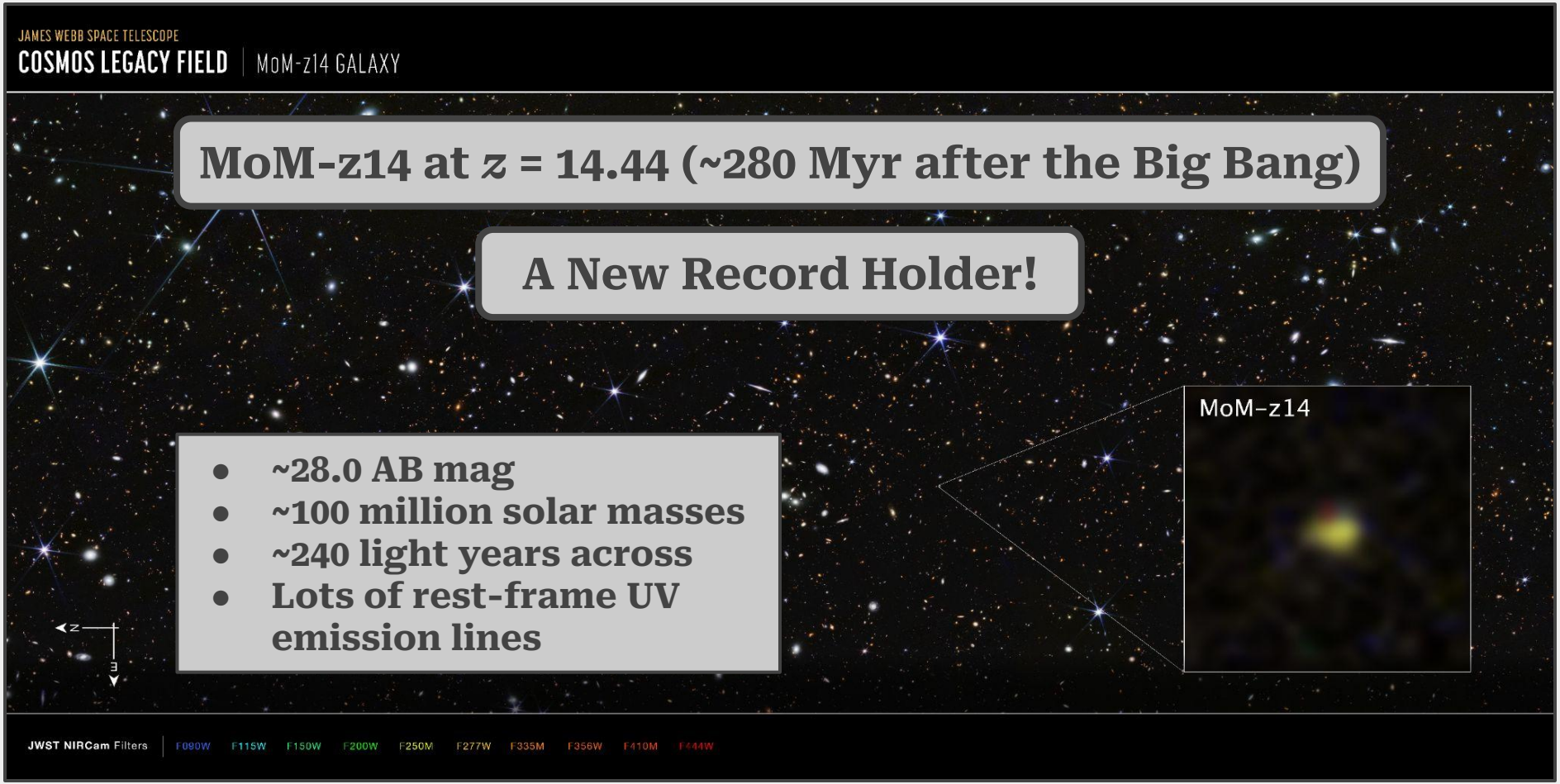


Image Credit: NASA, ESA, CSA, STScI, R. Naidu (MIT), J. DePasquale (STScI)

An Overabundance of Luminous Galaxies at $z > 10$? … Why?

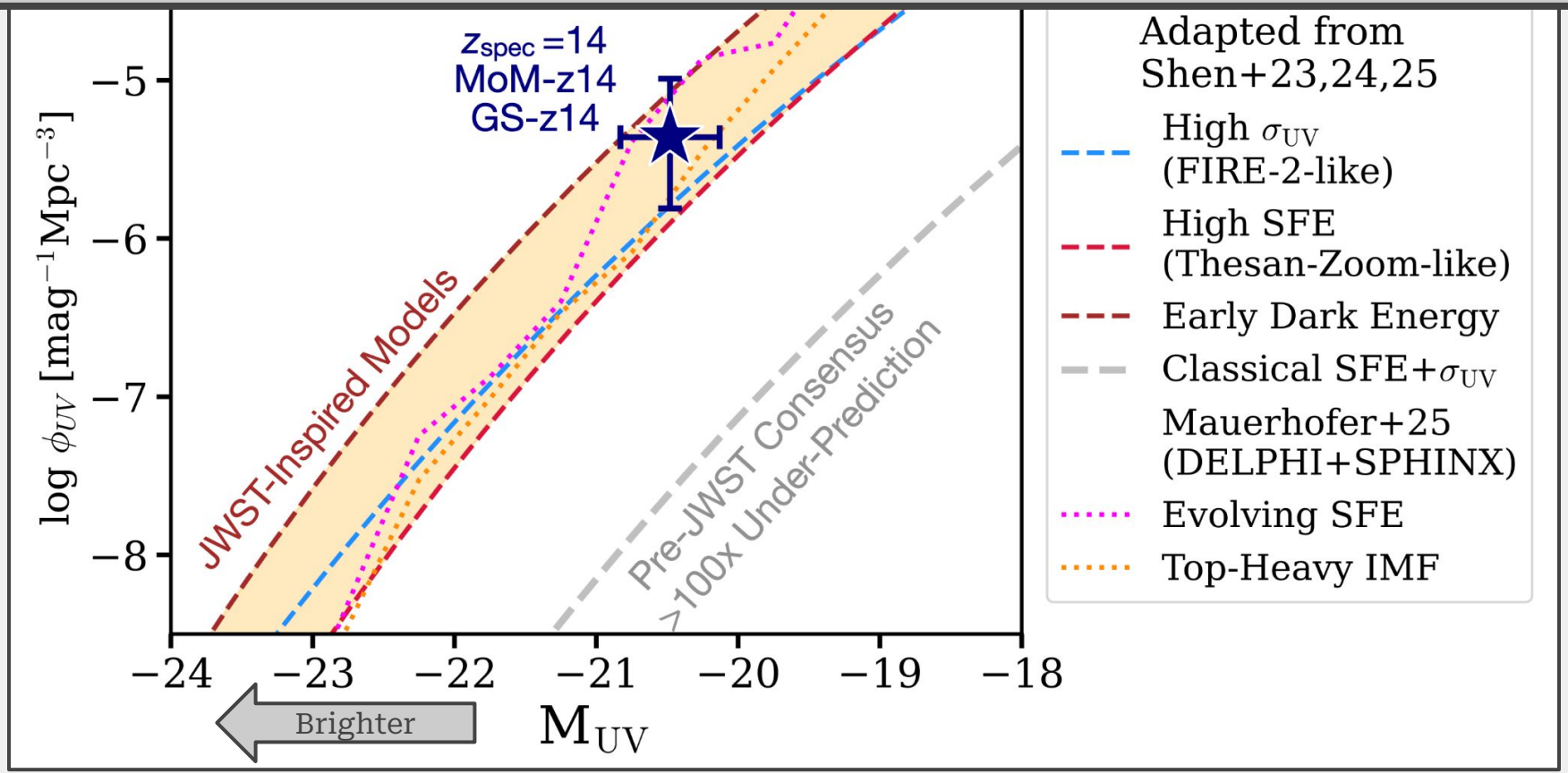


Image Credit: [Rohan P. Naidu et al. (2026)]

An Overabundance of Luminous Galaxies at $z > 10$? … Why?

#1. Bursty Star-Formation Histories (SFHs)

- Distant galaxies are predicted to experience ***intense episodic bursts of star formation***. During these bursts, the youngest and most massive stars will dominate the light and "outshine" the older stellar population that dominates the mass. ***Galaxies will appear more luminous during bursts of star formation, while obscuring evidence of prior star formation***.

#2. Increased Star Formation Efficiencies (SFEs)

- Distant galaxies may have been able to ***convert nebular gas into stars much more efficiently***. This increased star formation efficiency is possibly caused by feedback-free starbursts (FFBs), when the high densities and low metallicities at high redshifts guarantee ***increased SFEs in the most massive dark-matter halos due to ineffective feedback processes***.

#3. Different Stellar Initial Mass Functions (IMFs)

- Distant galaxies are expected to ***form stars differently than galaxies in the local Universe***, with IMFs that suppress low-mass star formation and enhance high-mass star formation due to lower metallicities and higher temperatures of the cosmic microwave background (CMB), ***resulting in brighter galaxies and accelerated metal enrichment at fixed halo mass***.

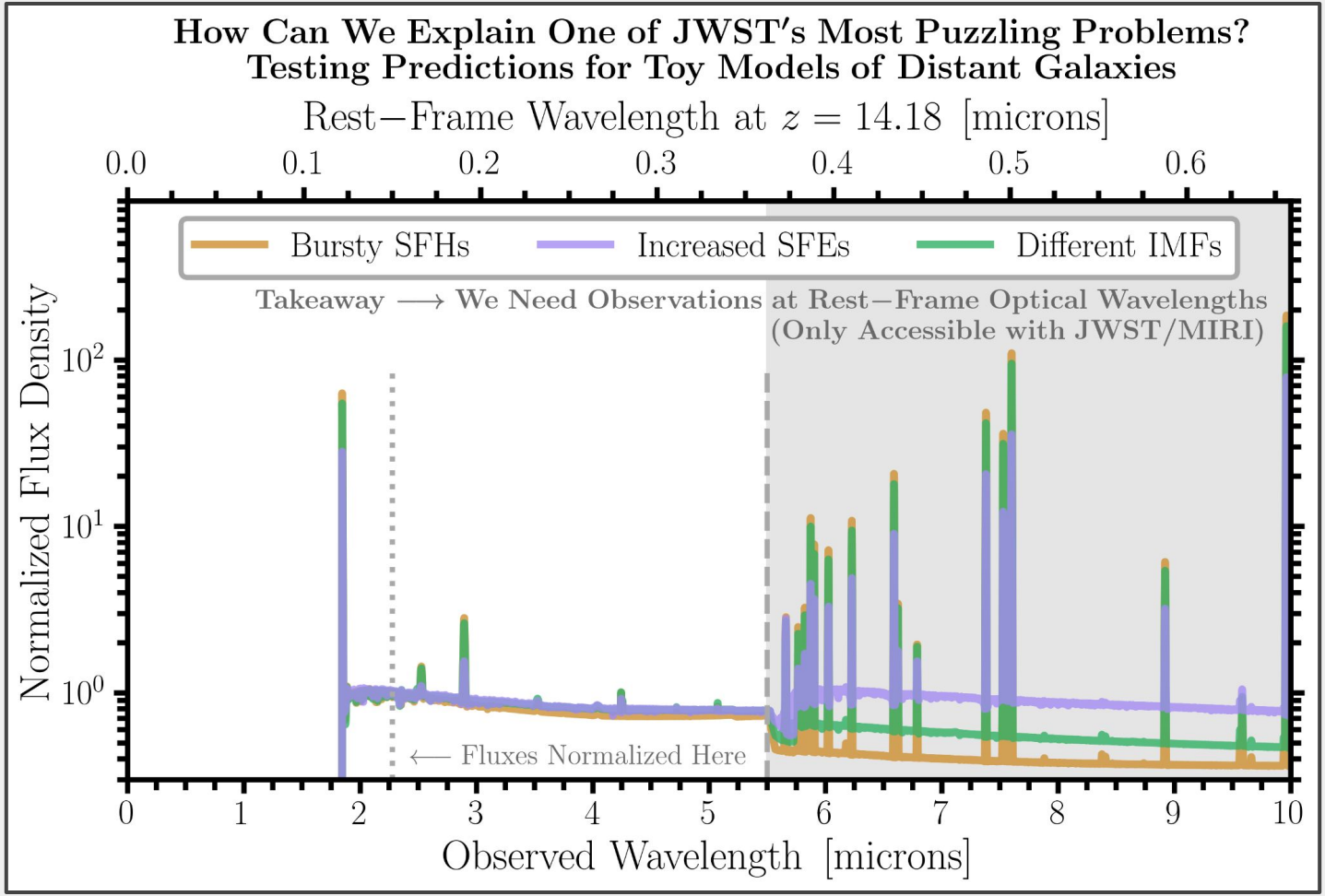


Image Credit: Jakob M. Helton

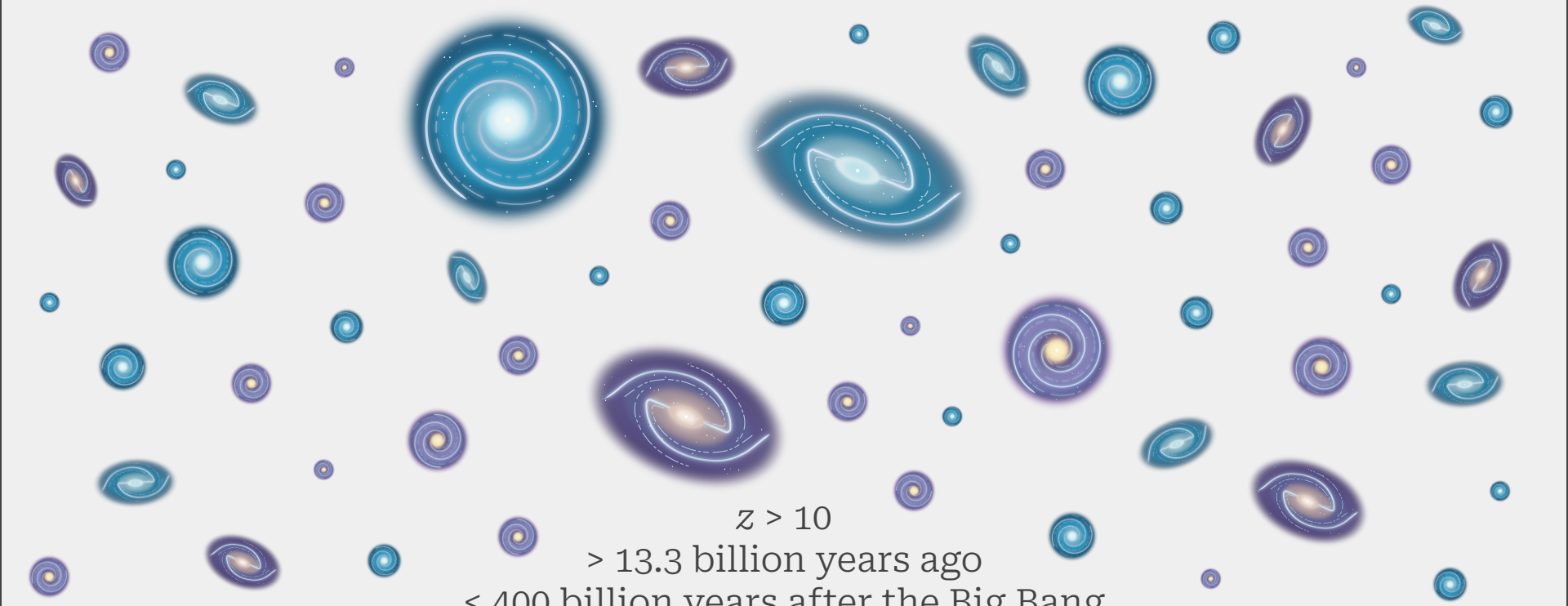
This is what we predicted to see with JWST...

$z > 10$
> 13.3 billion years ago
< 400 billion years after the Big Bang

Very few luminous galaxies at high redshifts, i.e., Cosmic Dawn

Image Credit: NASA

This is what we actually saw with JWST...



$z > 10$
> 13.3 billion years ago
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Lots of luminous galaxies at high redshifts, i.e., Cosmic Dawn

Image Credit: NASA

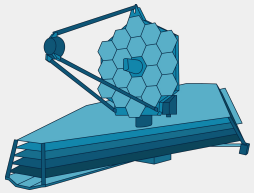
This is what we actually saw with JWST...

My Research Tries to Answer the Question of "Why?"

$z > 10$
> 13.3 billion years ago
< 400 billion years after the Big Bang

Lots of luminous galaxies at high redshifts, i.e., Cosmic Dawn

Image Credit: NASA



Relative Size on the Sky

JWST Advanced Deep Extragalactic Survey (JADES)

~770 hours of Observations!

JADES

30’

The Moon

~30% of the Moon’s Area

30’

Image Credit: [NASA, ESA, CSA, M. Zamani (ESA/Webb), the JADES Collaboration]

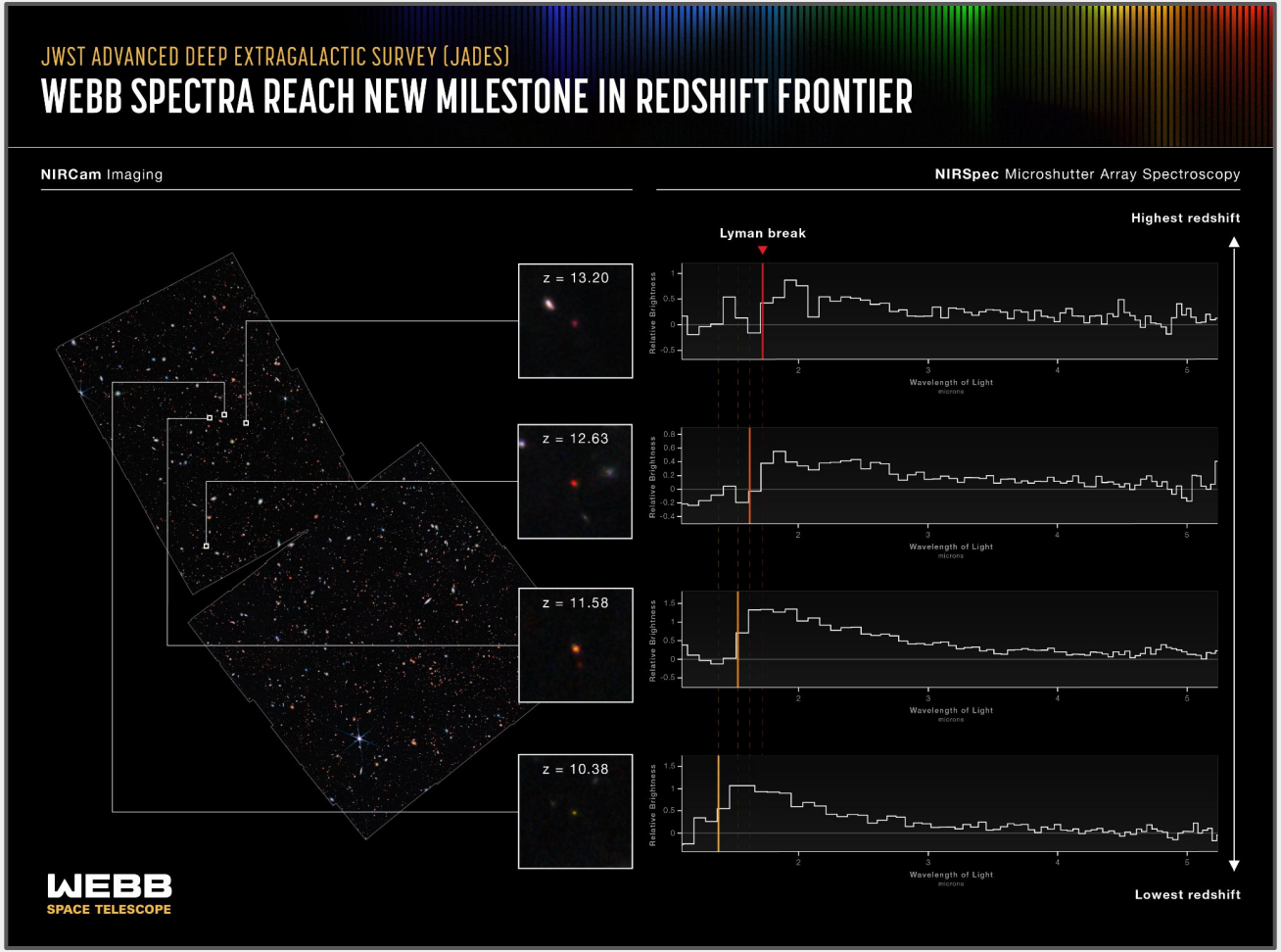
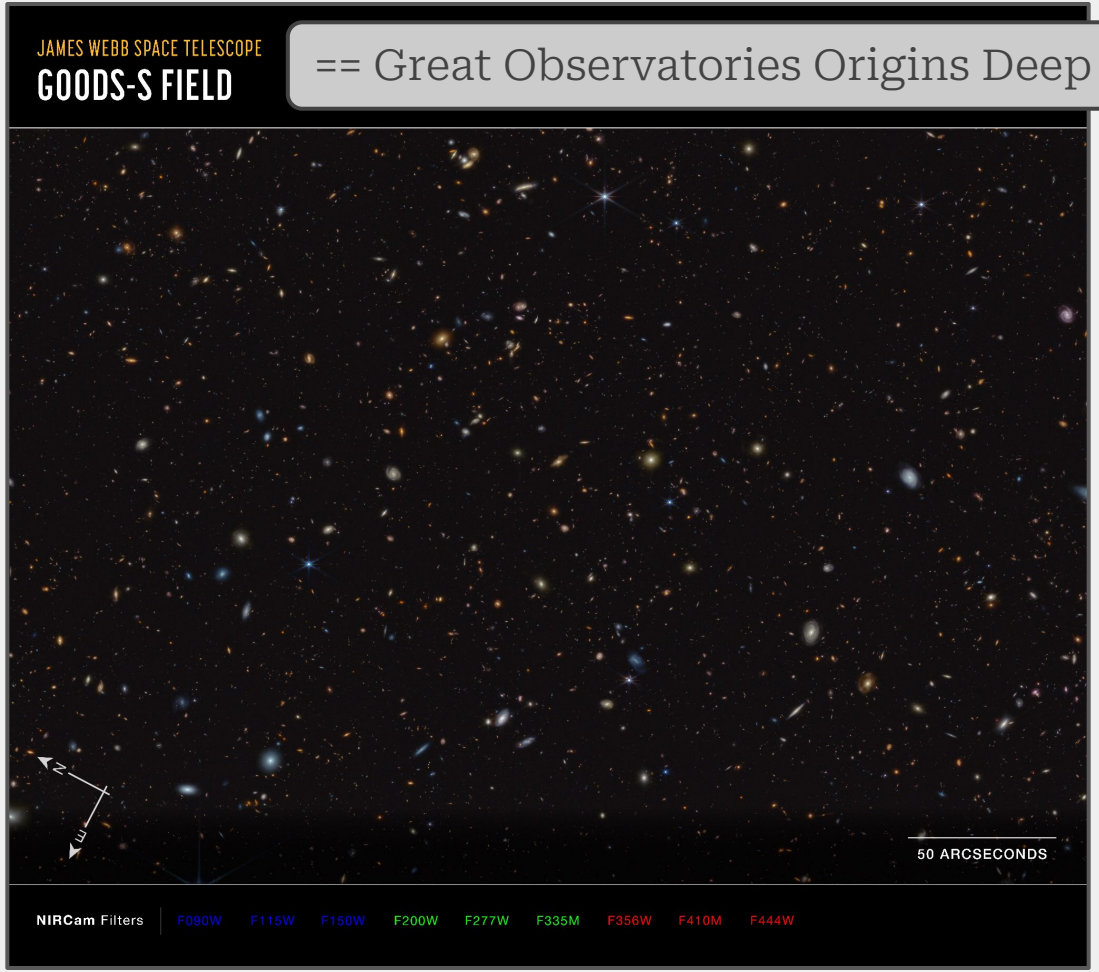


Image Credit: [NASA, ESA, CSA, M. Zamani (ESA), L. Hustak (STScI), the JADES Collaboration]



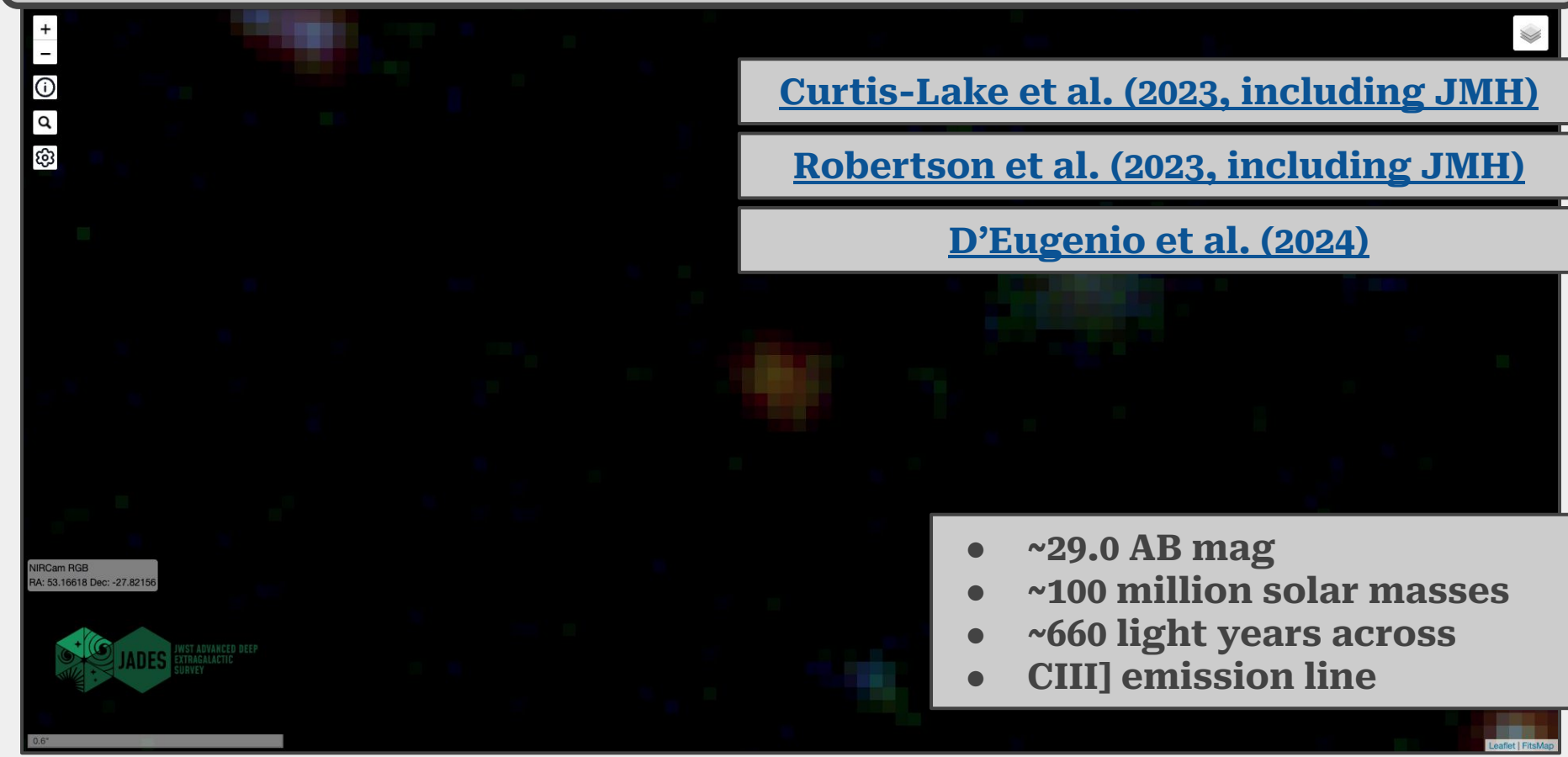
== Great Observatories Origins Deep Survey – South

Image Credit: [NASA, ESA, CSA, the JADES Collaboration, A. Pagan (STScI)]

Slide #33

Image Credit: NASA, ESA, CSA, the JADES Collaboration, A. Pagan (STScI)

JADES-GS-z12-0 at z = 12.48 (~345 Myr after the Big Bang) [Link]



Curtis-Lake et al. (2023, including JMH)

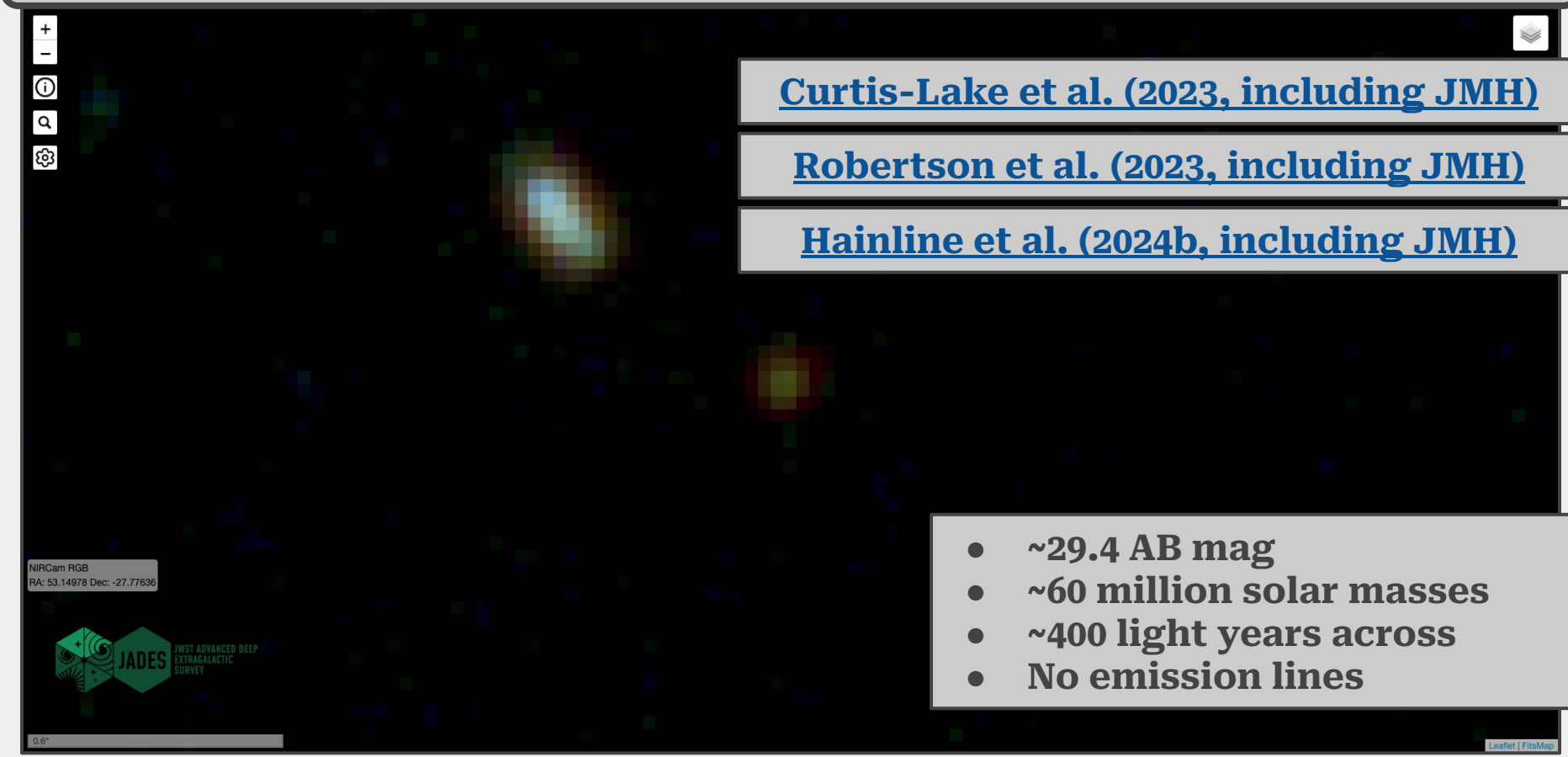
Robertson et al. (2023, including JMH)

D'Eugenio et al. (2024)

- ~29.0 AB mag
- ~100 million solar masses
- ~660 light years across
- CIII] emission line

Image Credit: The JADES Collaboration; https://jades.idies.jhu.edu/public/

JADES-GS-z13-0 at z = 13.16 (~320 Myr after the Big Bang) [Link]



Curtis-Lake et al. (2023, including JMH)

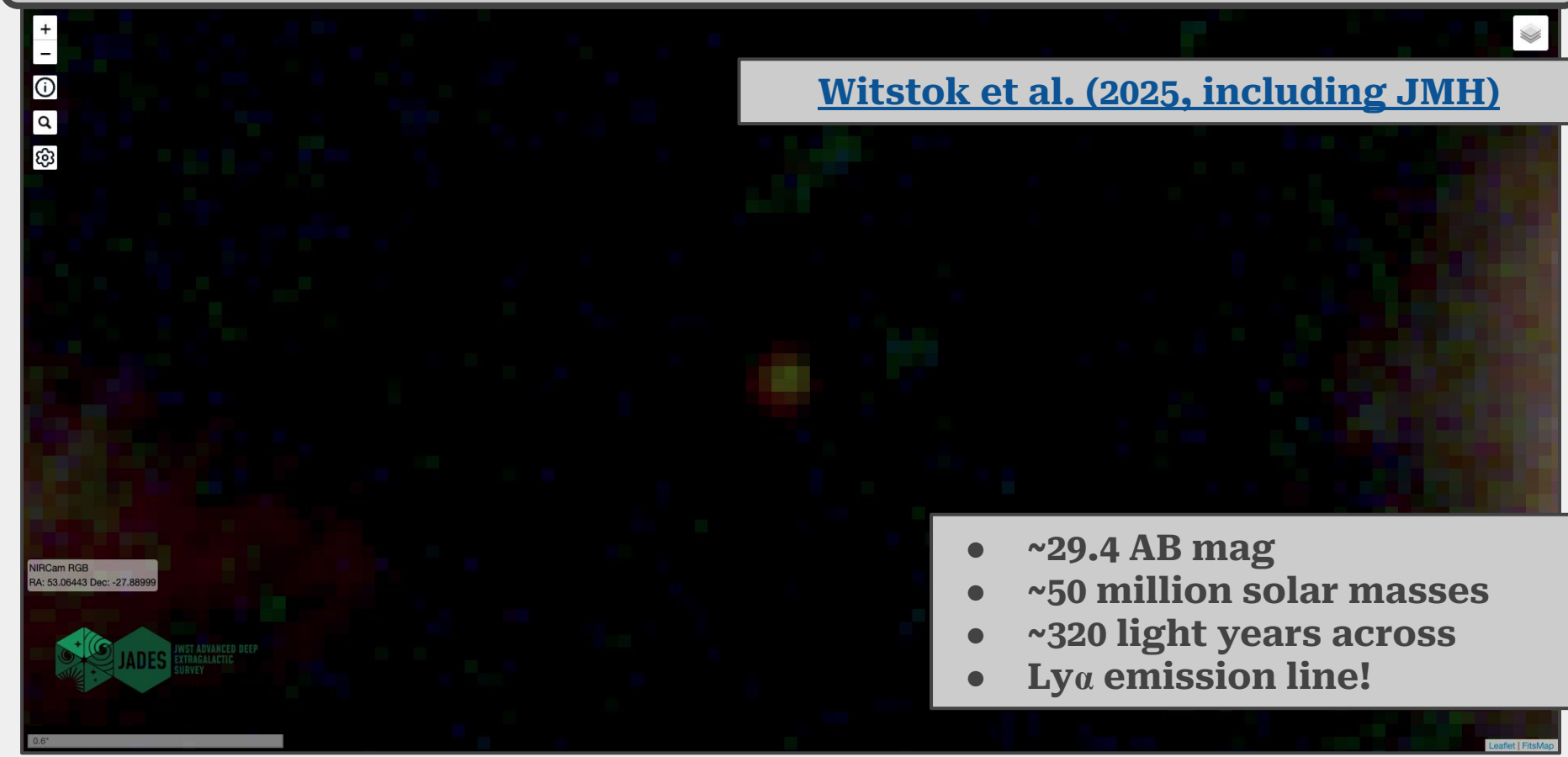
Robertson et al. (2023, including JMH)

Hainline et al. (2024b, including JMH)

- ~29.4 AB mag
- ~60 million solar masses
- ~400 light years across
- No emission lines

Image Credit: The JADES Collaboration; https://jades.idies.jhu.edu/public/

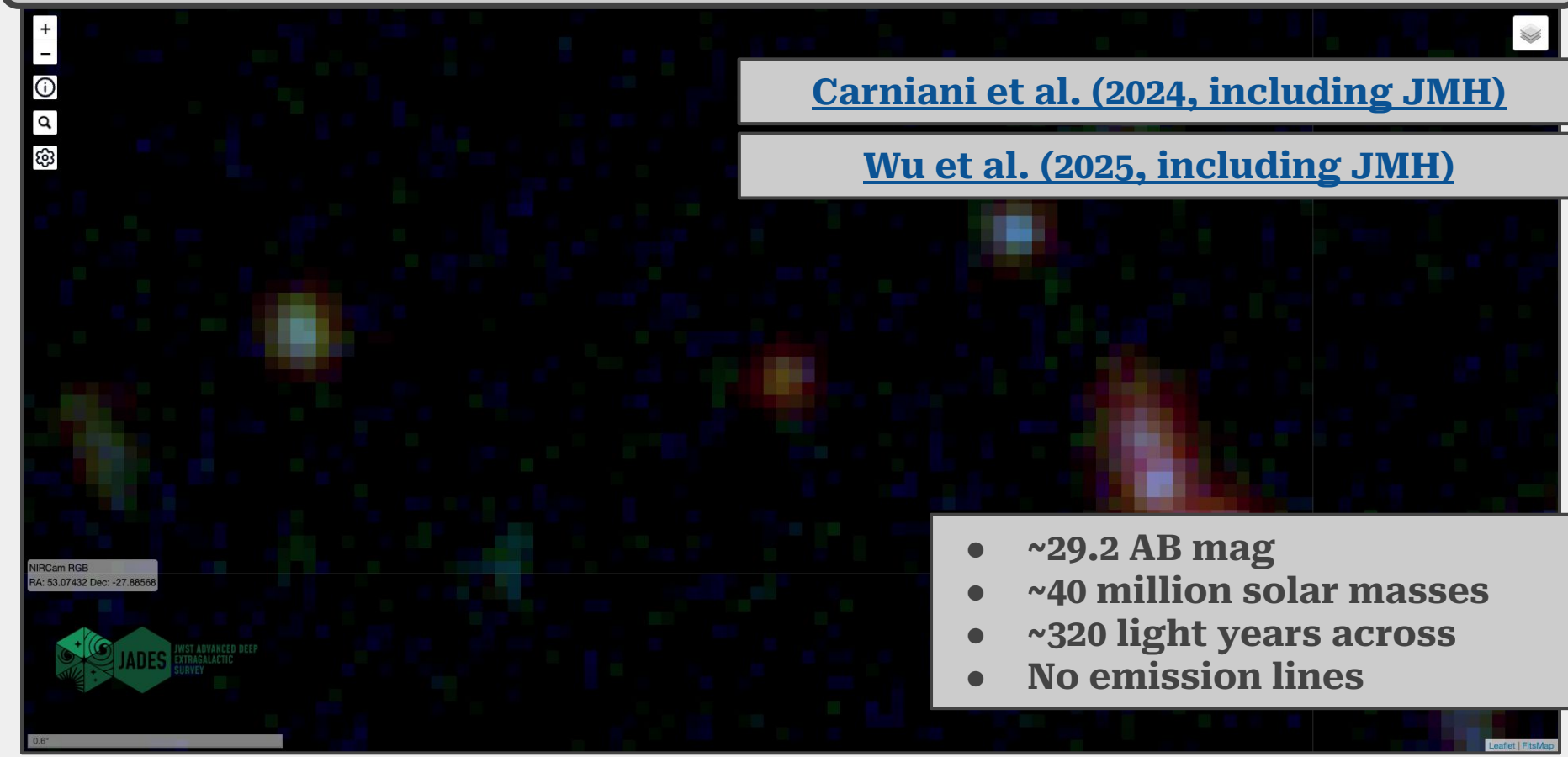
JADES-GS-z13-1 at z = 13.05 (~325 Myr after the Big Bang) [Link]



- ~29.4 AB mag
- ~50 million solar masses
- ~320 light years across
- Lyα emission line!

Image Credit: The JADES Collaboration; https://jades.idies.jhu.edu/public/

JADES-GS-z14-1 at z = 13.90 (~300 Myr after the Big Bang) [Link]



[Carniani et al. (2024, including JMH)]

[Wu et al. (2025, including JMH)]

- ~29.2 AB mag
- ~40 million solar masses
- ~320 light years across
- No emission lines

Image Credit: [The JADES Collaboration; https://jades.idies.jhu.edu/public/]

JADES-GS-z14-0 at z = 14.18 (~290 Myr after the Big Bang) [Link]

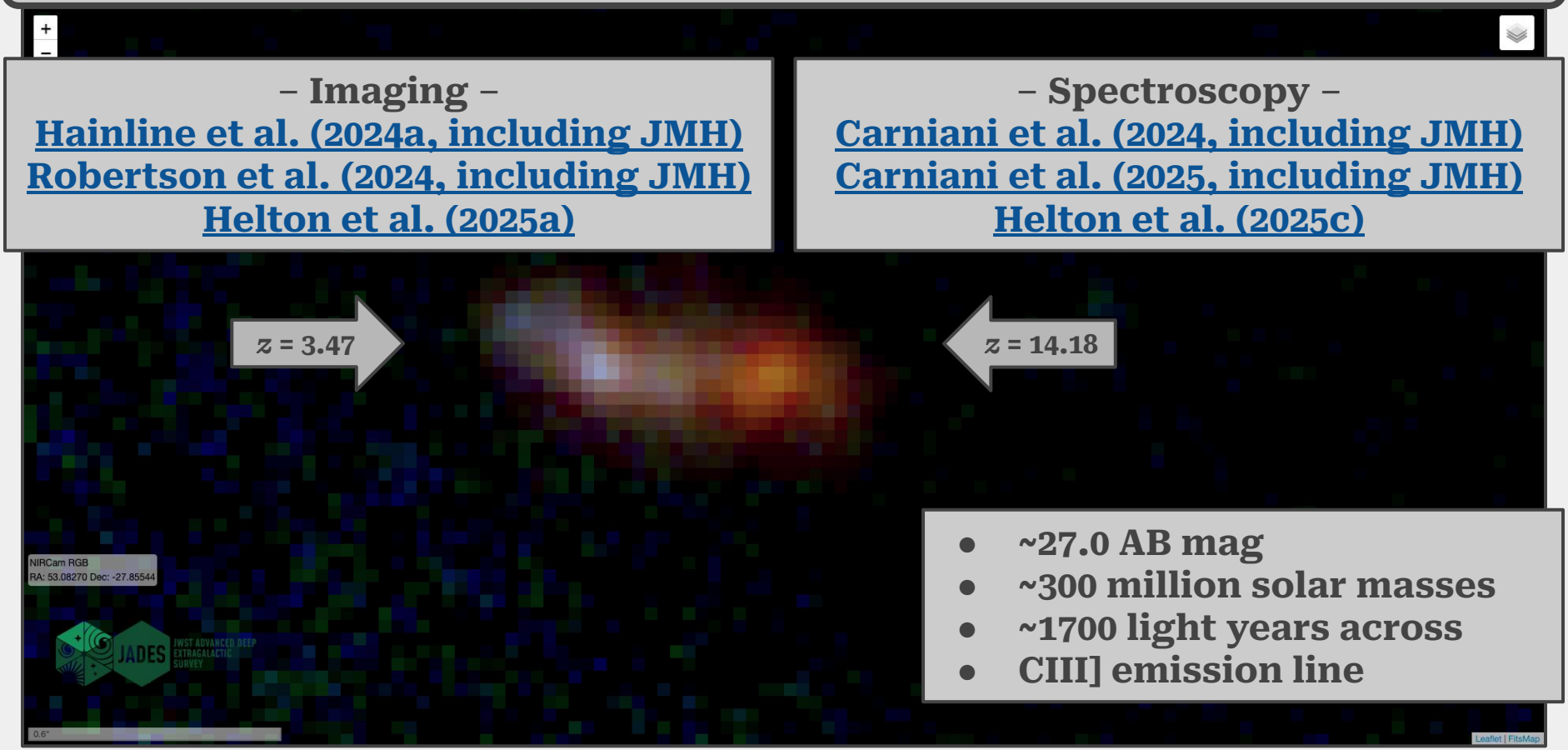
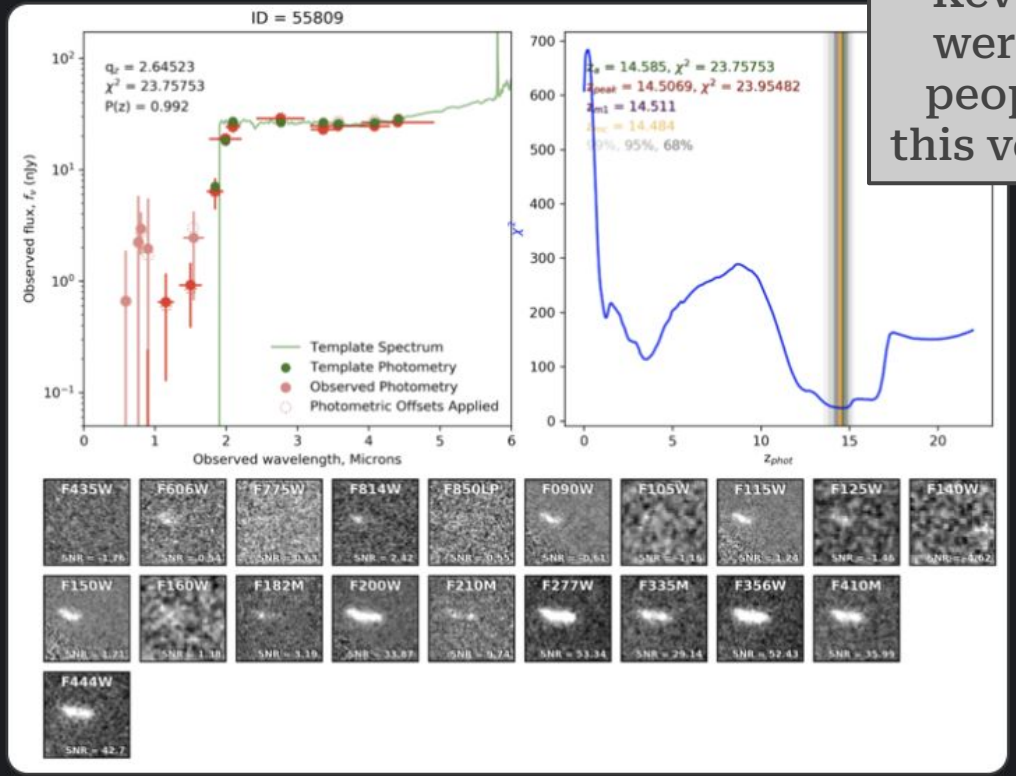


Image Credit: The JADES Collaboration; https://jades.idies.jhu.edu/public/

this is a true weirdo

image.png



The first ever message about JADES-GS-z14-0!

Kevin Hainline and I were two of the first people to lay eyes on this very distant galaxy.

Image Credit: Jakob M. Helton

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JWST Detects the Earliest, Most Distant Galaxy in the Known Universe—And It's Super Weird

NASA's James Webb Space Telescope observed an unusually large and highly luminous galaxy at a record-breaking 290 million years after the big bang

BY RIIS WILLIAMS EDITED BY LEE BILLINGS

Artist's illustration of NASA's James Webb Space Telescope. James Vaughan/Science Photo Library/Alamy Stock Photo

Image Credit: Scientific American

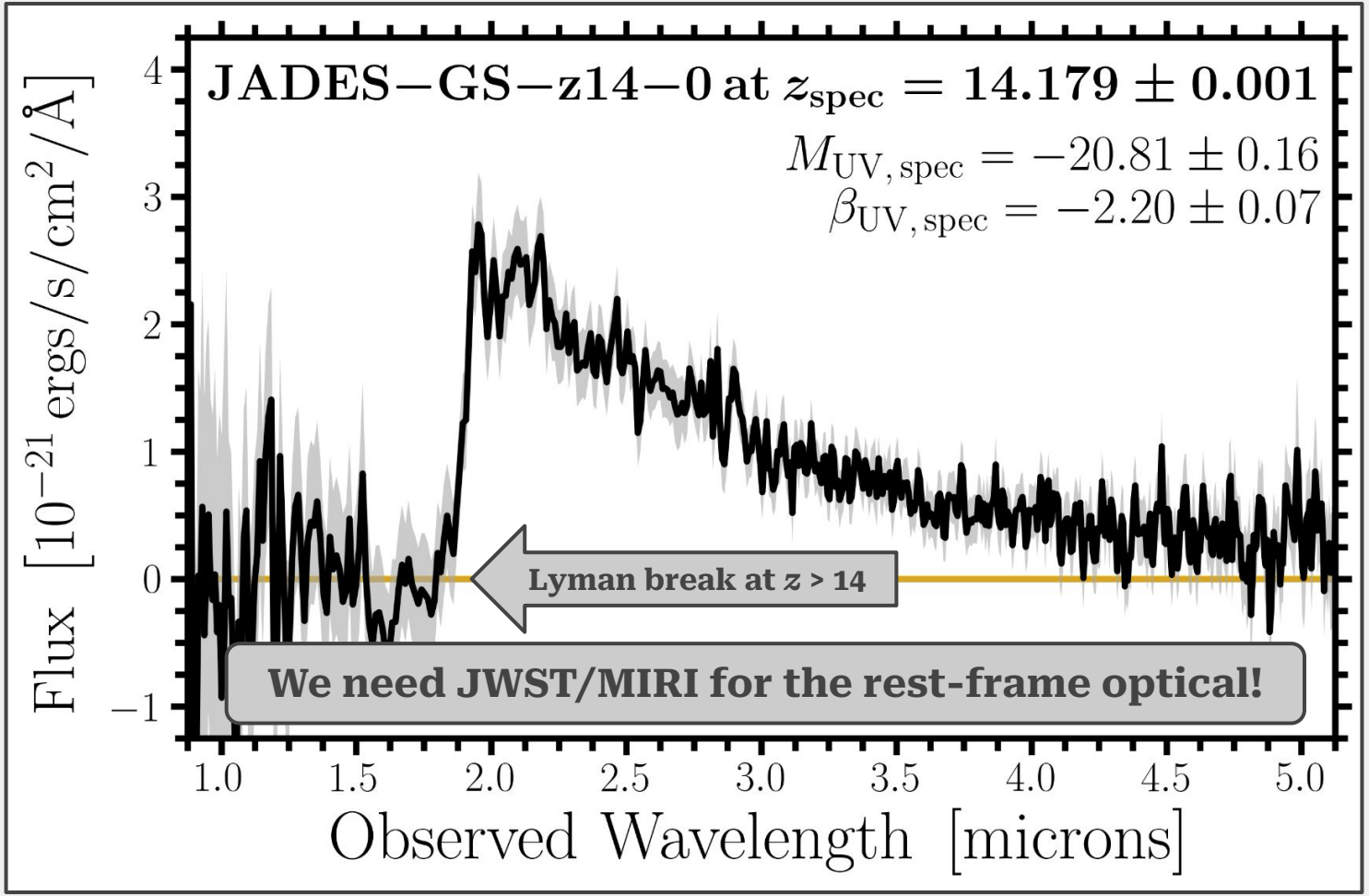
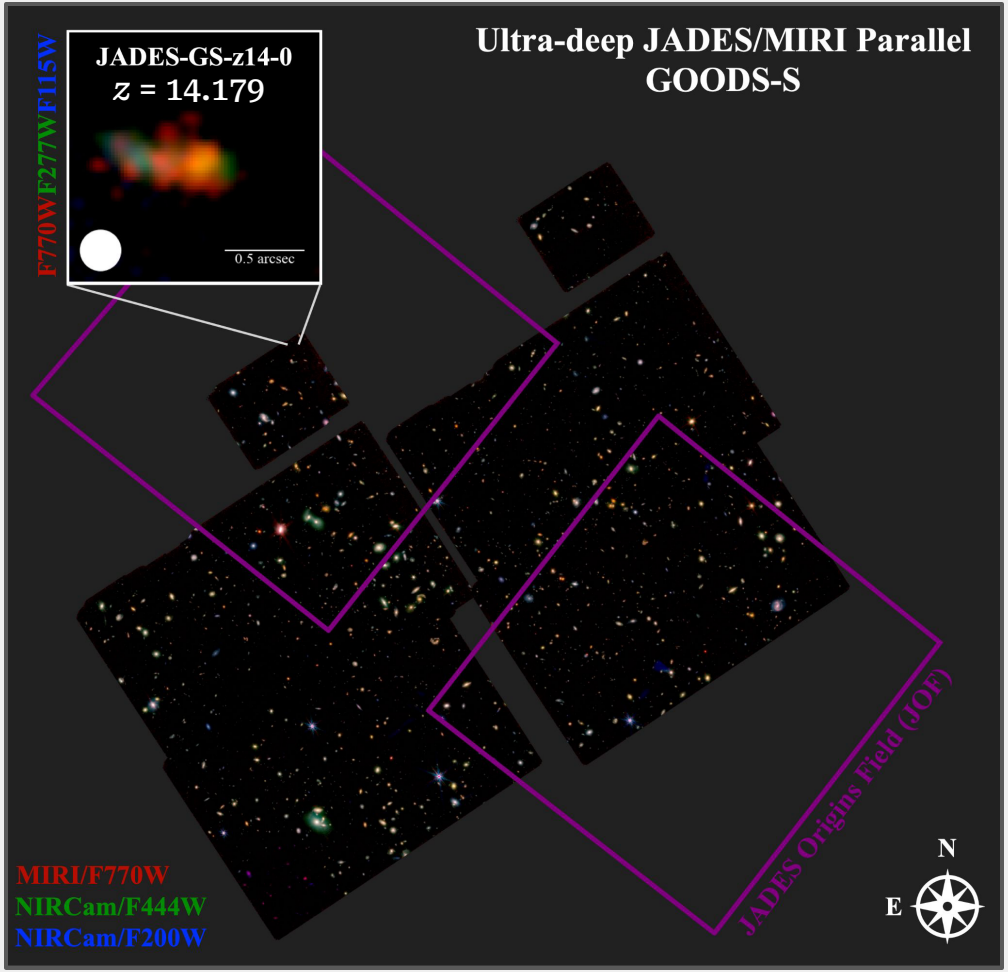


Image Credit: Jakob M. Helton

Total on-source integration time is ~43 hours with MIRI/Imaging



Total on-source integration time is ~167 hours with NIRCam/Imaging

Image Credit: Stacey Alberts

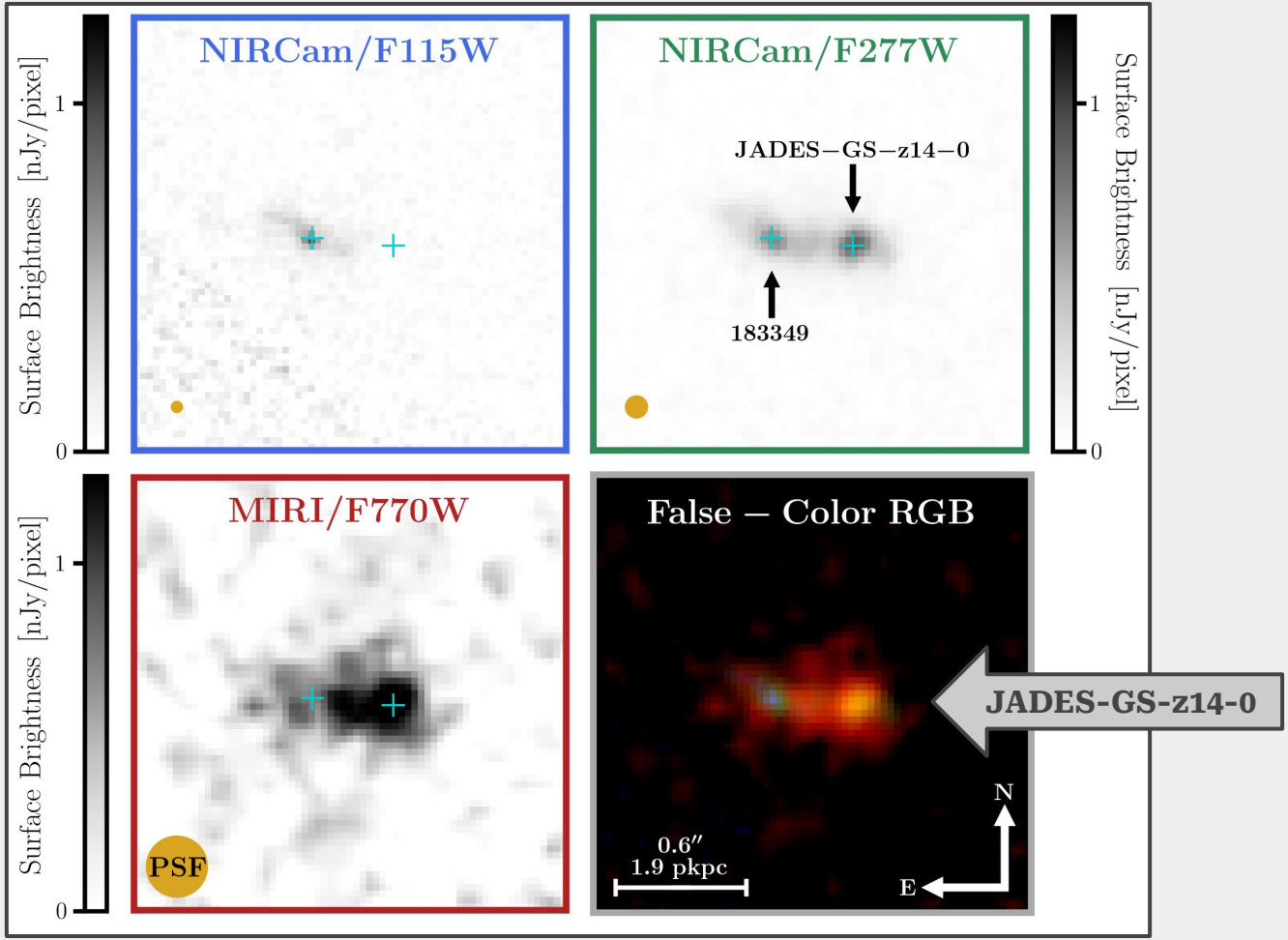


Image Credit: [Jakob M. Helton et al. (2025a)]

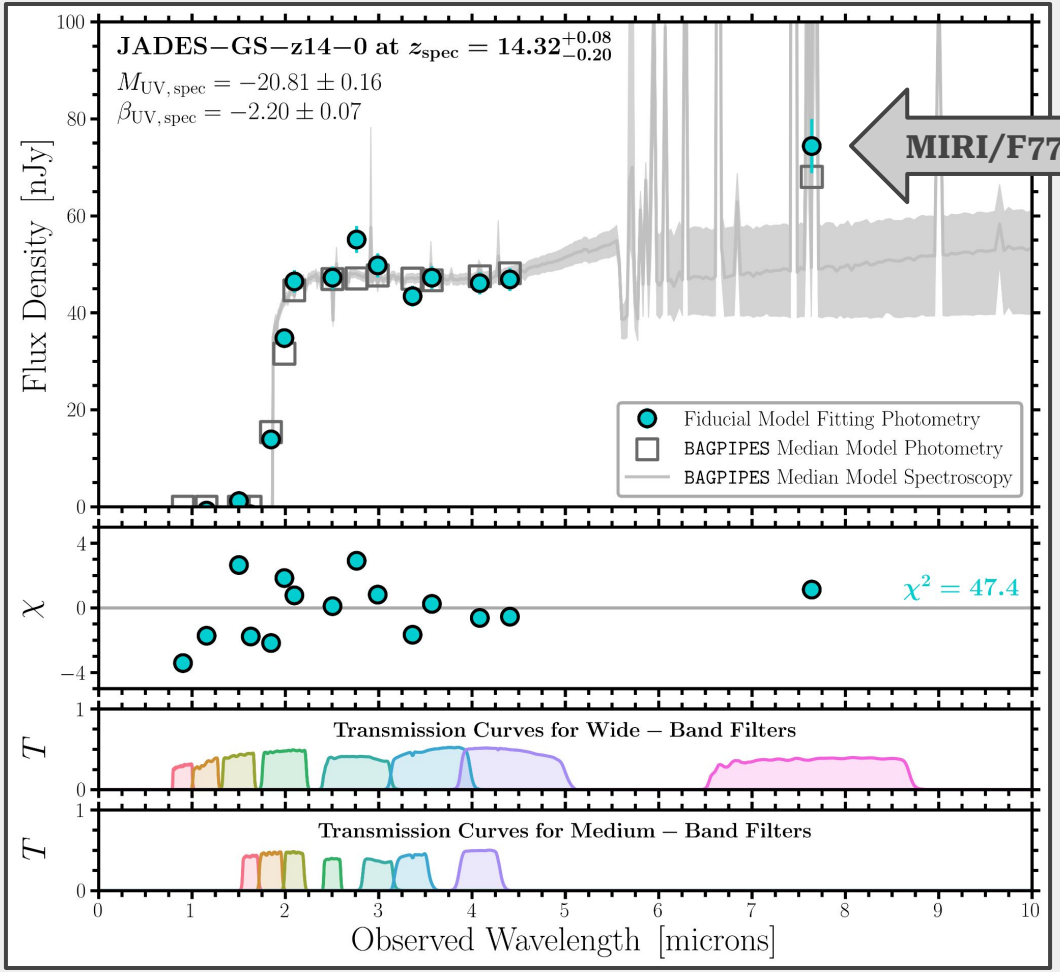


Image Credit: Jakob M. Helton et al. (2025a)

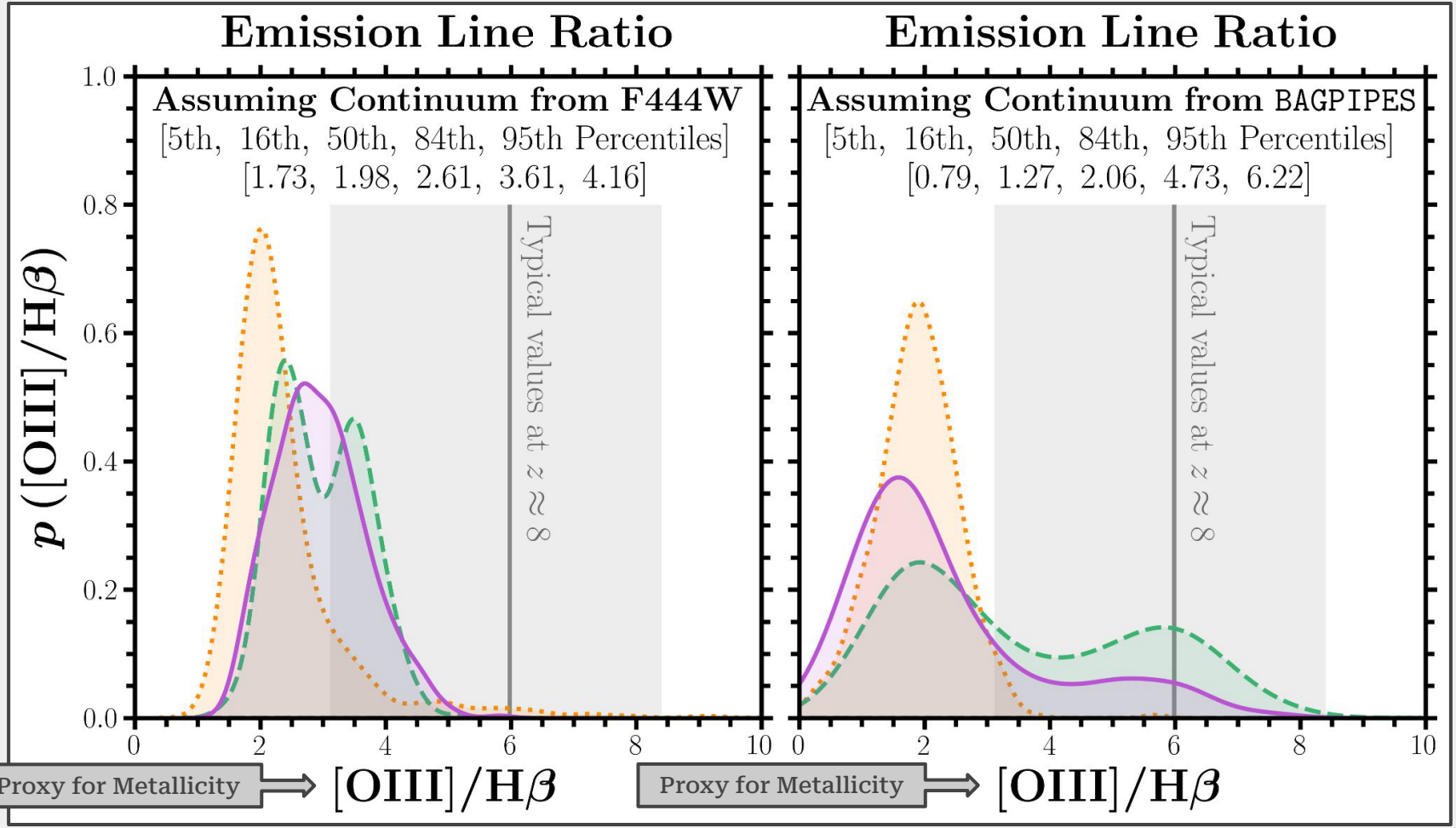


Image Credit: [Jakob M. Helton et al. (2025a)]

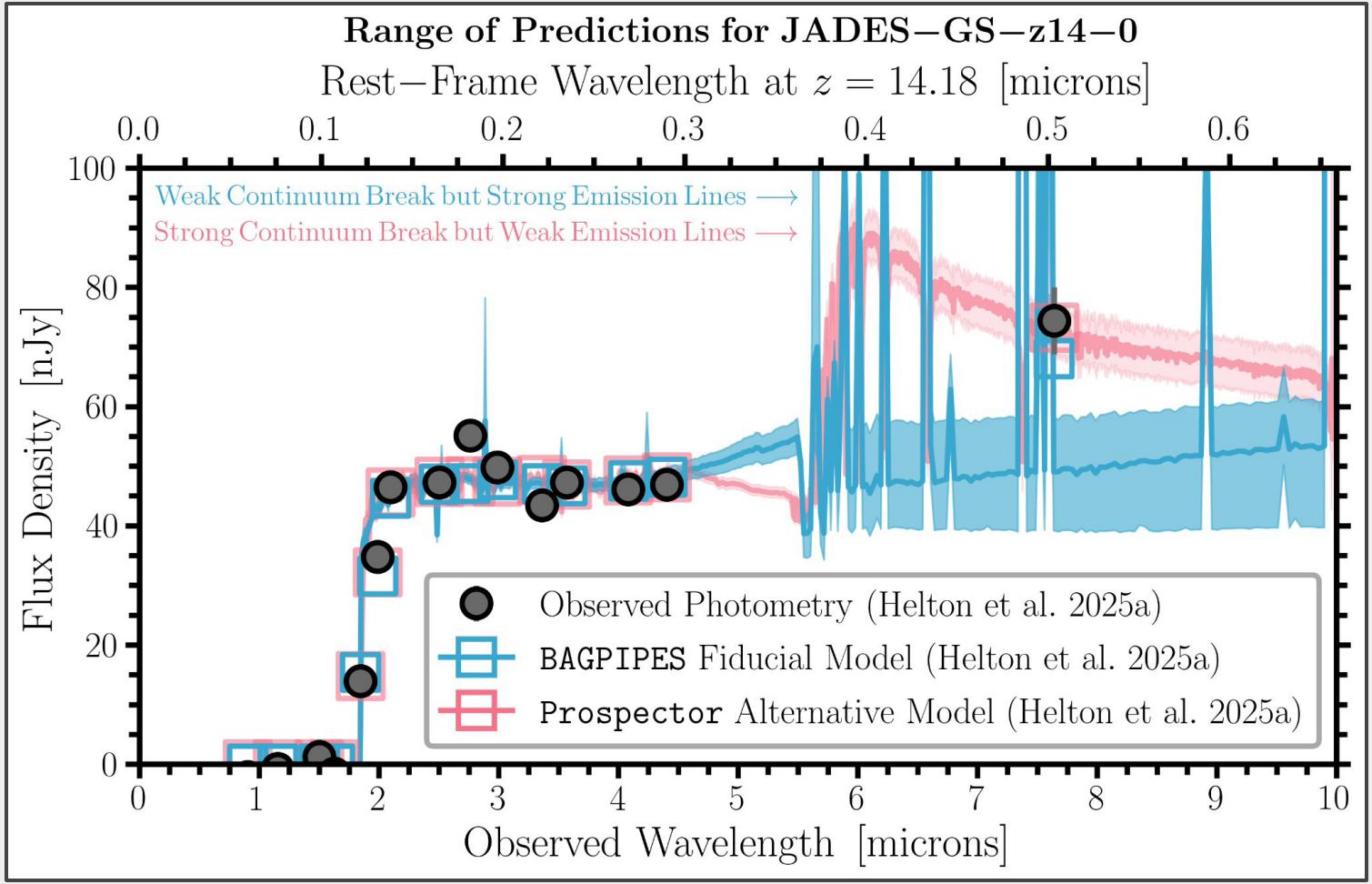
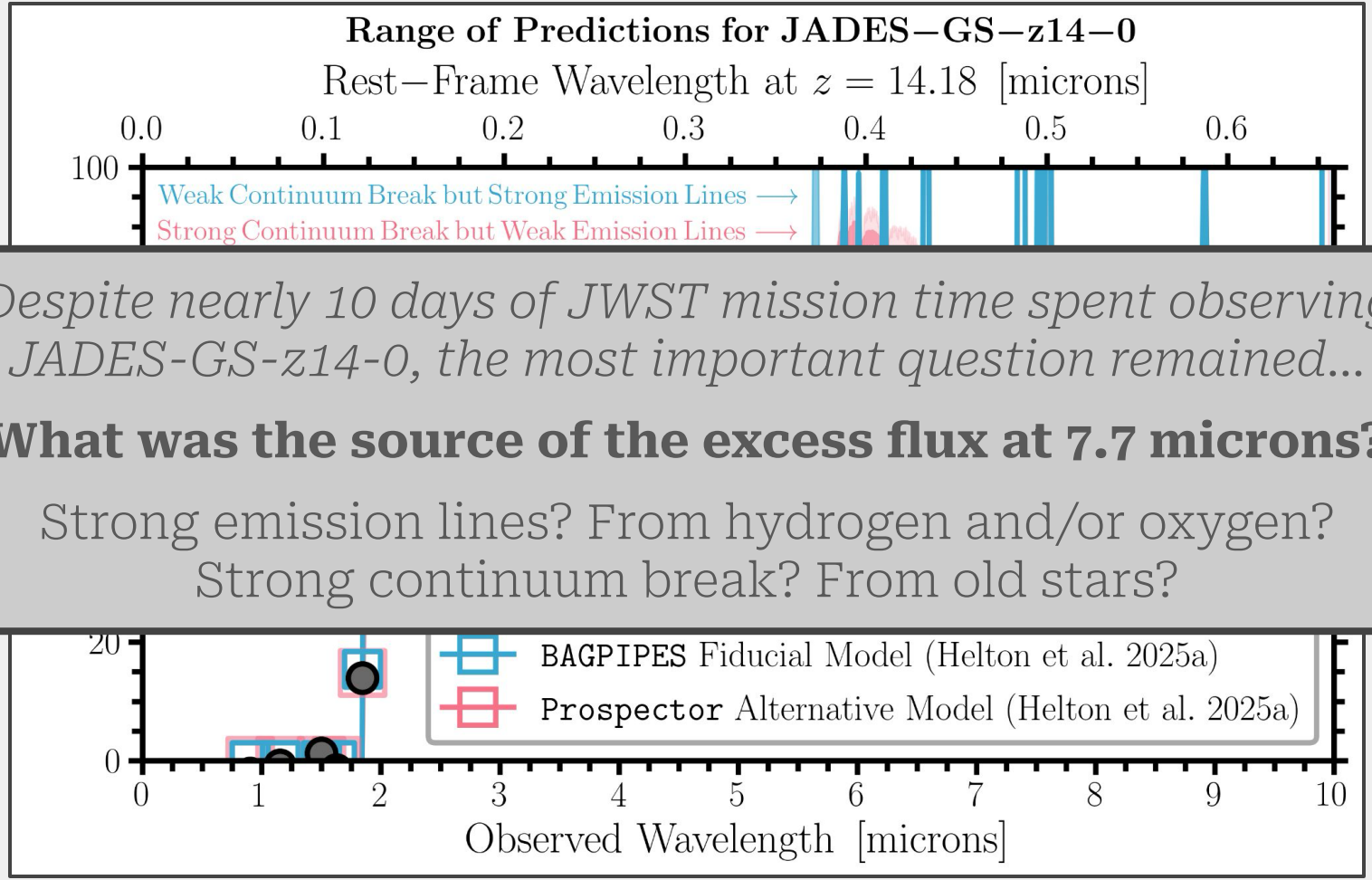


Image Credit: Jakob M. Helton



Despite nearly 10 days of JWST mission time spent observing JADES-GS-z14-0, the most important question remained...

What was the source of the excess flux at 7.7 microns?

Strong emission lines? From hydrogen and/or oxygen?
Strong continuum break? From old stars?

Image Credit: Jakob M. Helton

ALMA Observations of the [OIII] Line at 88 microns

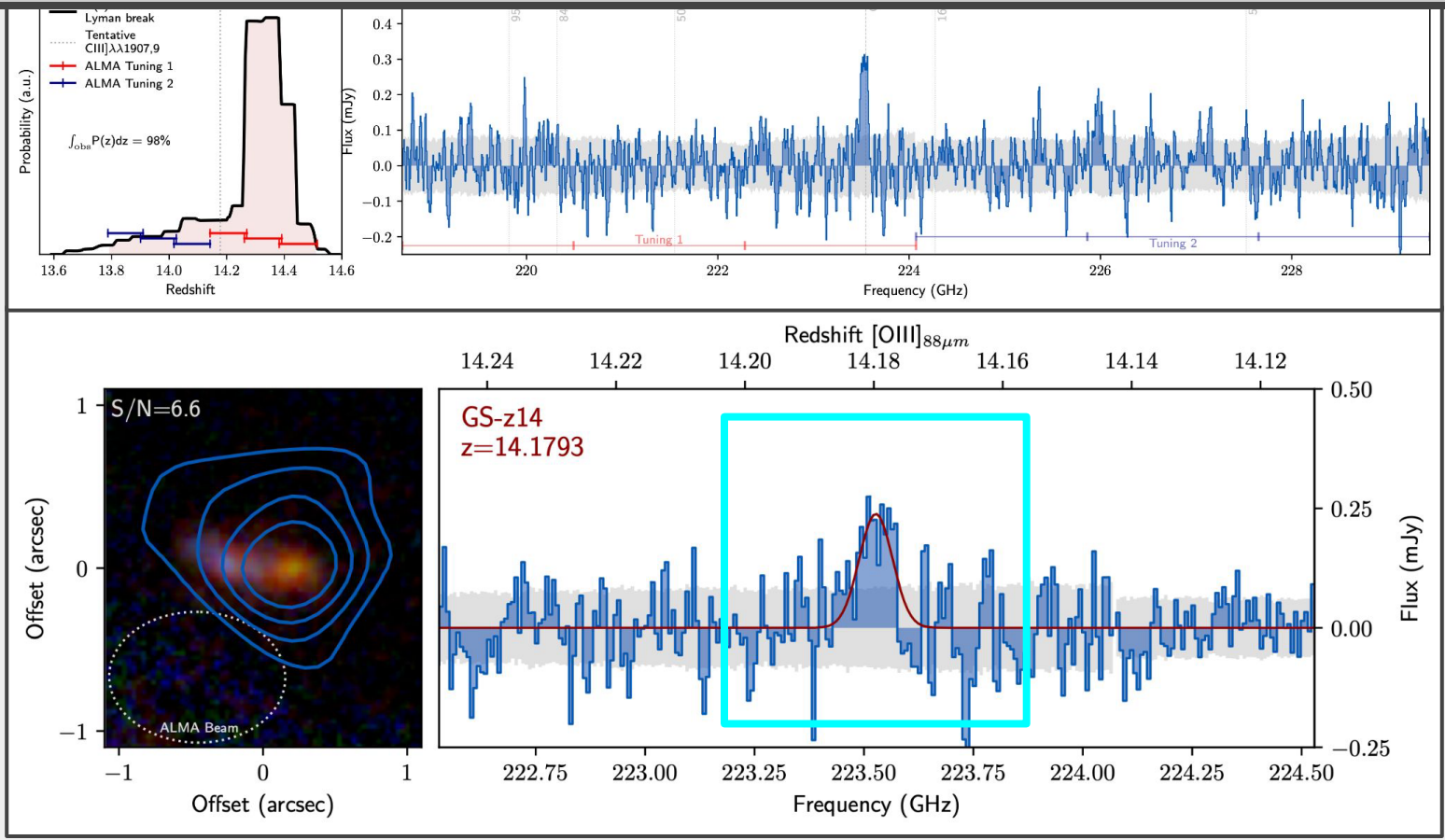


Image Credit: Sander Schouws et al. (2025)

My successful JWST Cycle 4 proposal!

James Webb Space Telescope Cycle 4 GO Proposal **8544**

Rest-Frame Optical Nebular Emission Lines at Cosmic Dawn: MIRI/LRS Follow-Up for JADES-GS-z14-0

Scientific Category:	High-redshift Galaxies and the Distant Universe
Scientific Keywords:	Cosmic dawn, Emission line galaxies, High-redshift galaxies
Instruments:	MIRI
Proposal Size:	Medium
Exclusive Access Period:	0 months (less than default of 12 months)
Allocation Information (in hours):	
Science Time:	48.2
Charged Time:	62.6

Abstract

The recent spectroscopic confirmation of JADES-GS-z14-0 at z = 14.179 (Carniani et al. 2024a, Carniani et al. 2024b, Schouws etal. 2024) challenges our understanding of the early Universe. This galaxy is remarkably luminous (M_UV = -20.81) and reddened (beta_UV = -2.20). It is clearly spatially resolved in the rest-frame ultraviolet (R_UV = 260 parsecs) and therefore dominated by stellar light at those wavelengths. Most notably, it has been detected at 7.7 microns with JWST/MIRI, and the measured flux density in MIRI/F770W (74.4 nJy, S/N = 13) is in excess above the measured flux density with JWST/NIRCam (Helton et al. 2024). At this redshift, less than 300 million years after the Big Bang, the measured flux excess is likely caused by nebular emission from hydrogen and ionized oxygen. We propose for ultra-deep spectroscopic follow-up observations with the MIRI/LRS to cover the brightest rest-frame optical nebular emission lines ([OII]3727+3729, Hbeta, [OIII]4959+5007, and Halpha) in addition to the stellar continuum. These observations will unveil important insights into this extreme galaxy: (1) gas-phase metallicity, electron density, and ionization parameter; (2) recent star-formation rate, ionizing photon production efficiency, and diffuse dust attenuation; and (3) properties of the stellar populations, including stellar mass, stellar age, and shape of the star-formation history. JADES-GS-z14-0 represents the current archetype of early galaxy formation, and the proposed observations represent a rare opportunity to study the formation of some of the first stars and heavy elements in the Universe.

Observations were acquired in November of 2025!

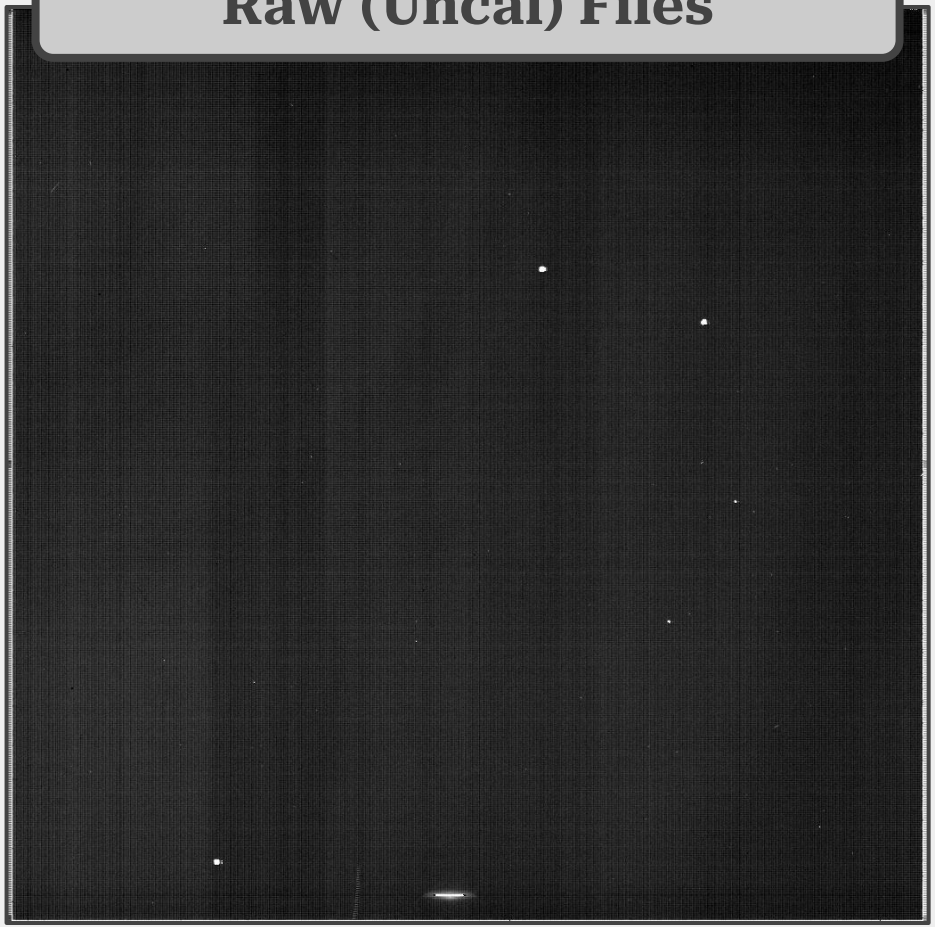
Summary of Challenges with Data Processing

- **We have to convert > 200 GB of images into one < 200 KB spectrum.**
- **At least 50% of the pixels in each image are affected by cosmic ray events.**
- The emission line measurements are highly sensitive to assumptions about the local continuum, wavelength calibration, and wavelength-dependent resolution.
- The uncertainties reported by the pipeline are underestimated by ~60-65%.
- The final spectrum is contaminated by a neighboring foreground galaxy.
- The default pipeline does a not-so-great job of reducing the data.
- Plus many other subtle detector artifacts and slit loss effects...

– <u>Takeaway</u> –

JWST/MIRI is arguably the least understood instrument on board JWST; the MIRI/LRS spectrum that I obtained is the longest and most technically challenging observing program ever conducted by JWST/MIRI.

Raw (Uncal) Files



Partially Processed Files

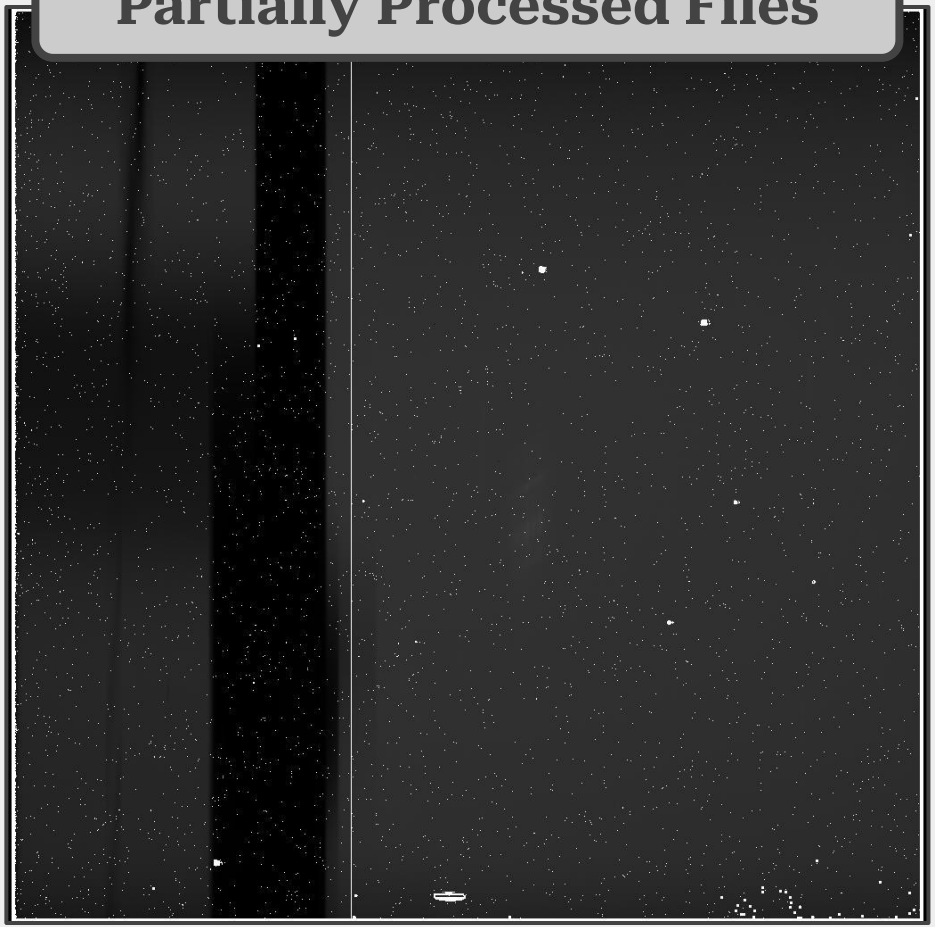


Image Credit: Jakob M. Helton

GOES Proton Flux (i.e., Cosmic Rays) from NOAA

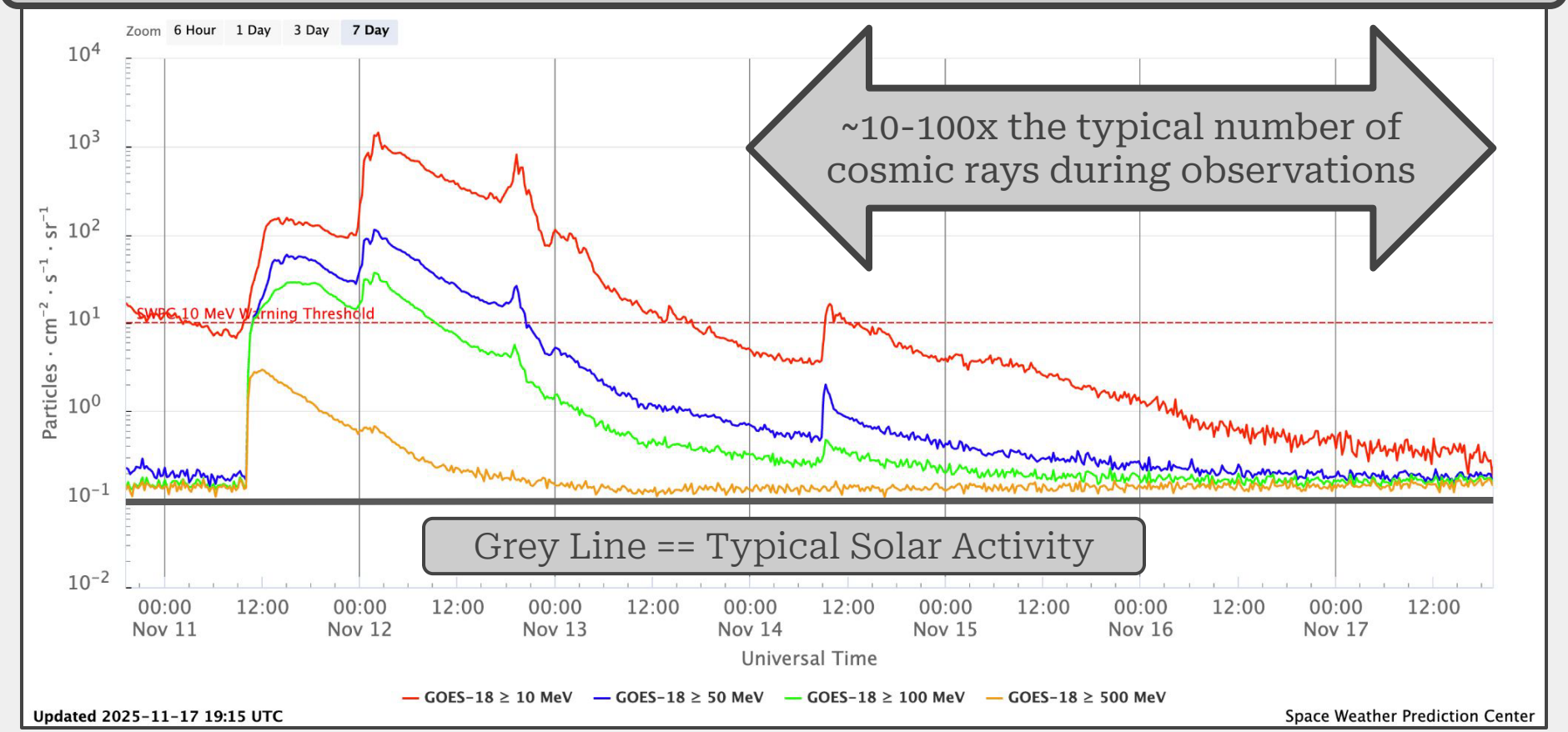
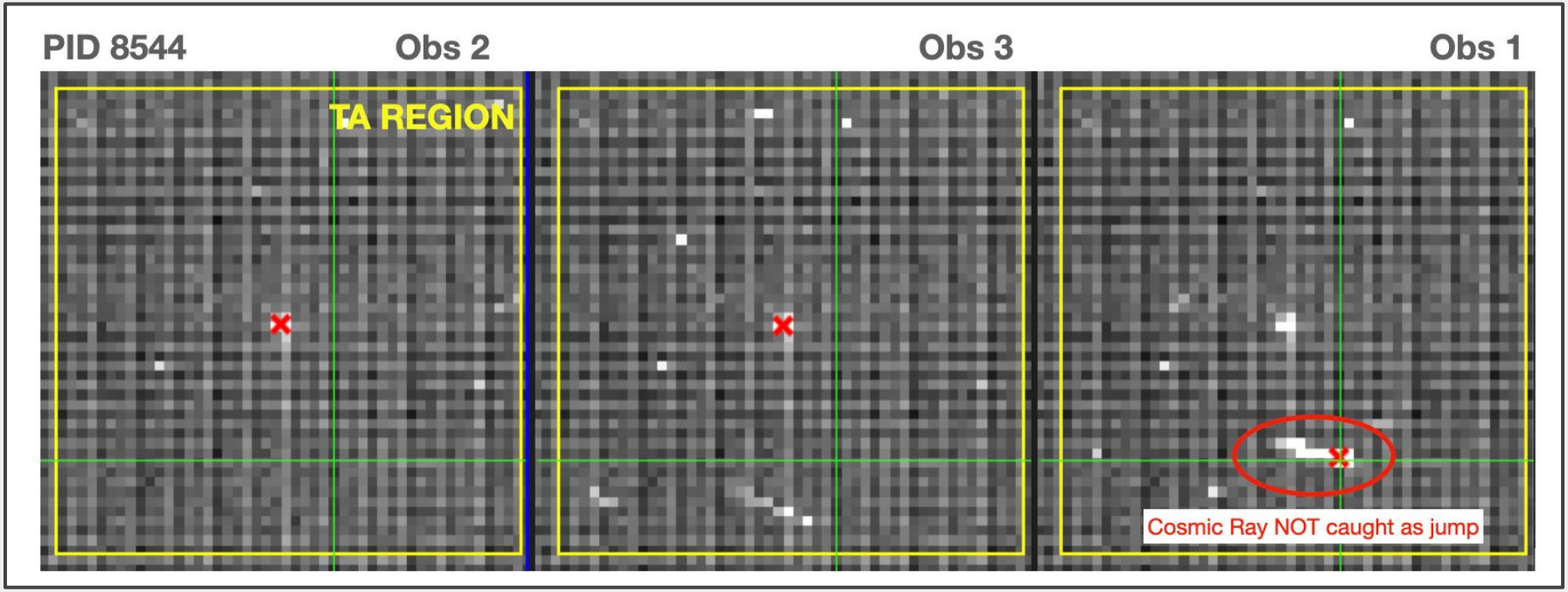


Image Credit: NOAA

The elevated solar activity ruined 1/3 of our observations!



JWST performed target acquisition (TA) on a cosmic ray...

Image Credit: Stacey Alberts

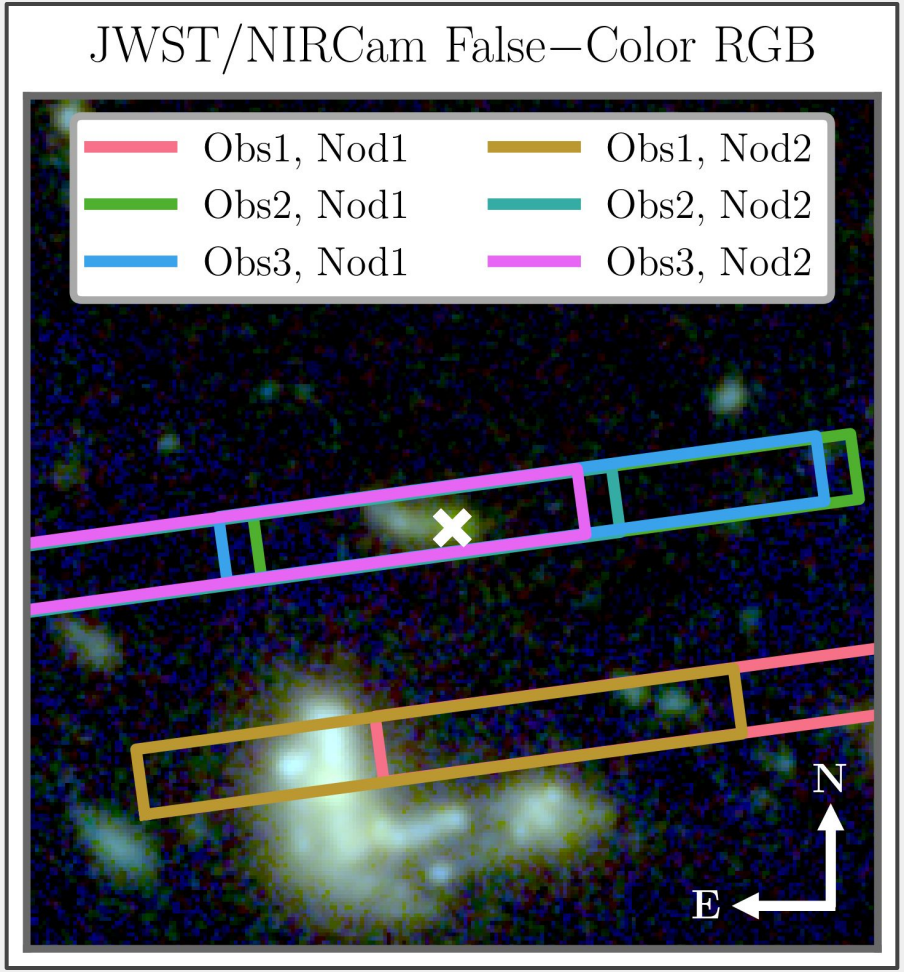


Image Credit: [Jakob M. Helton et al. (2025c)]

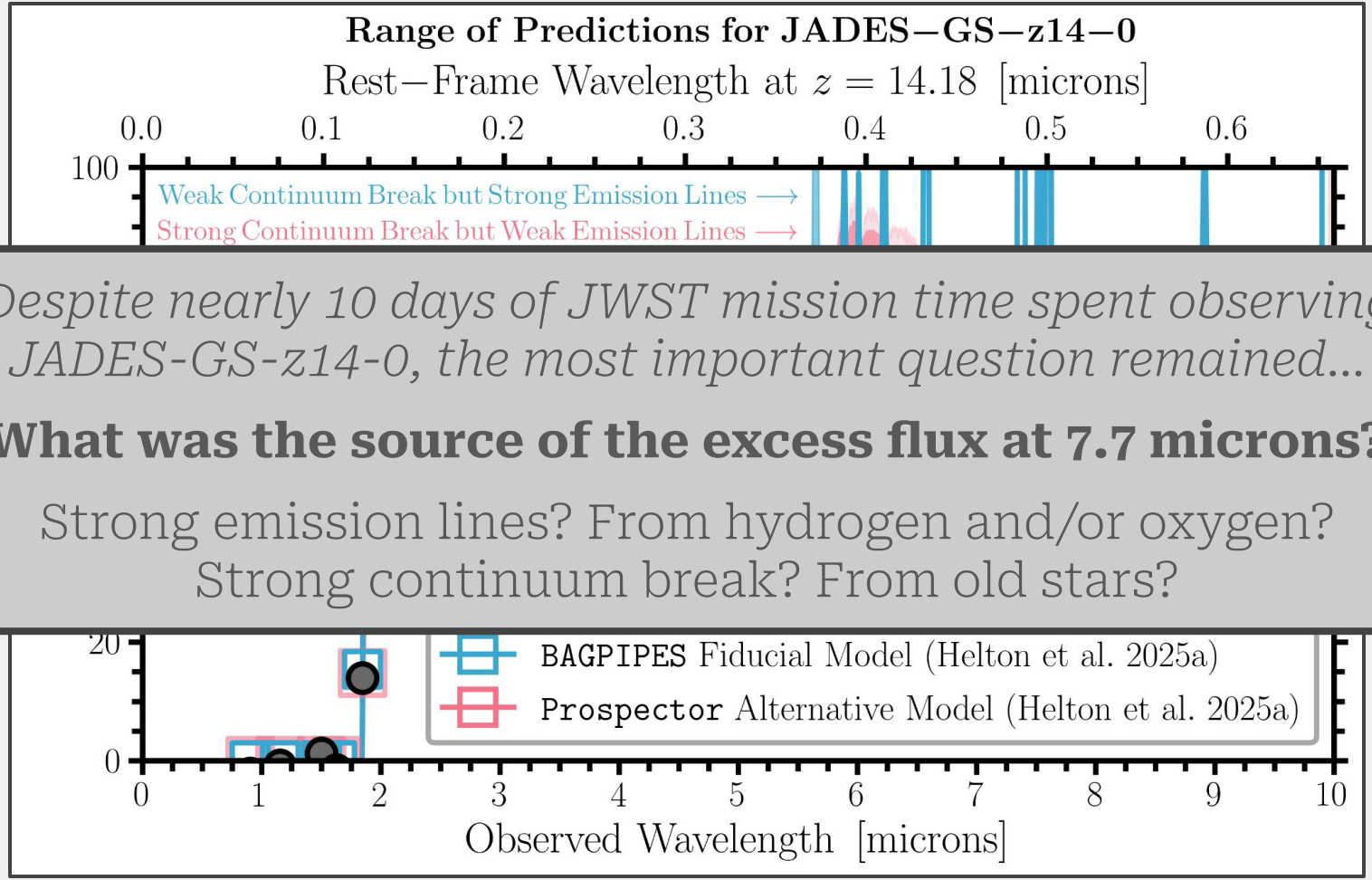


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Image Credit: Jakob M. Helton

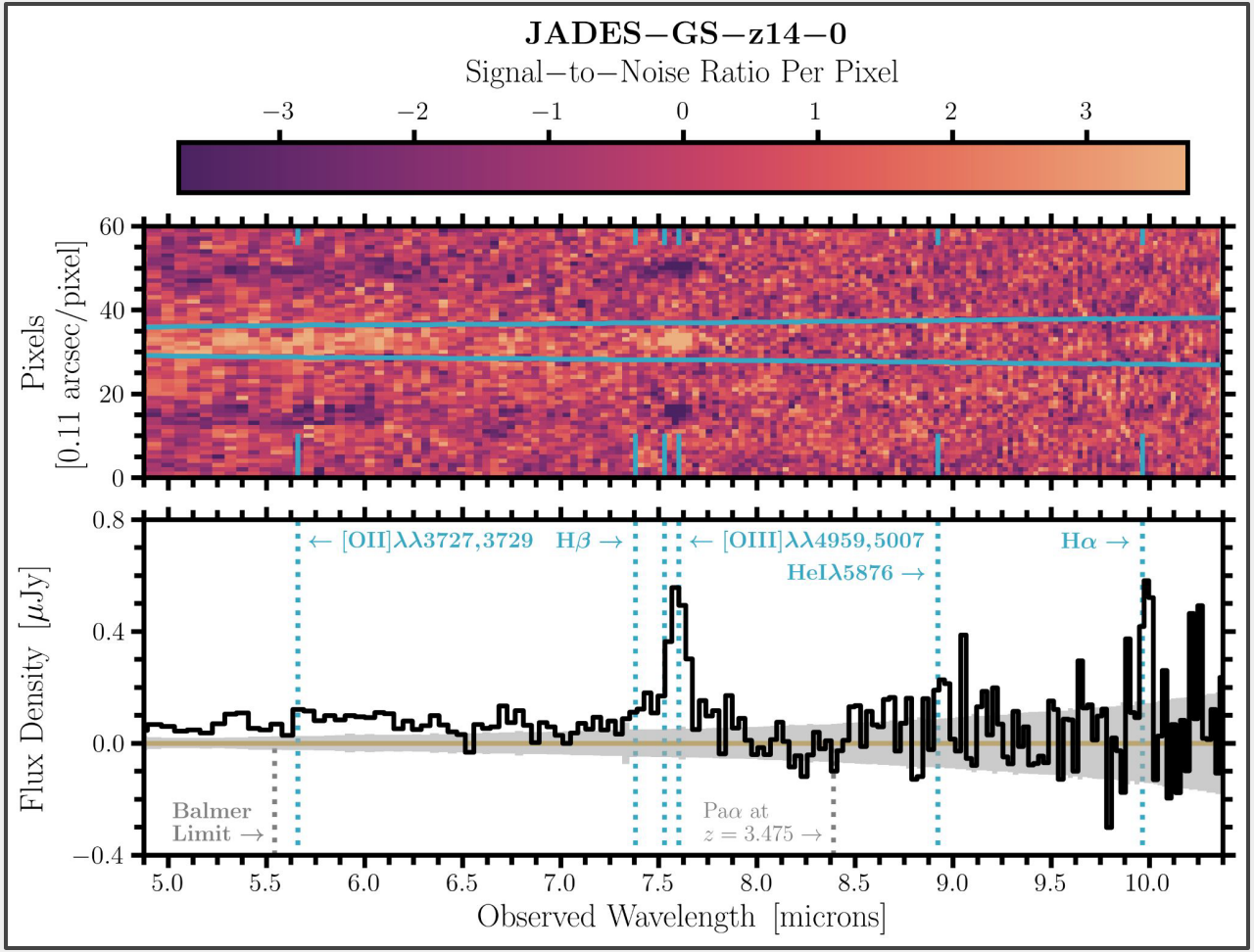


Image Credit: Jakob M. Helton et al. (2025c)

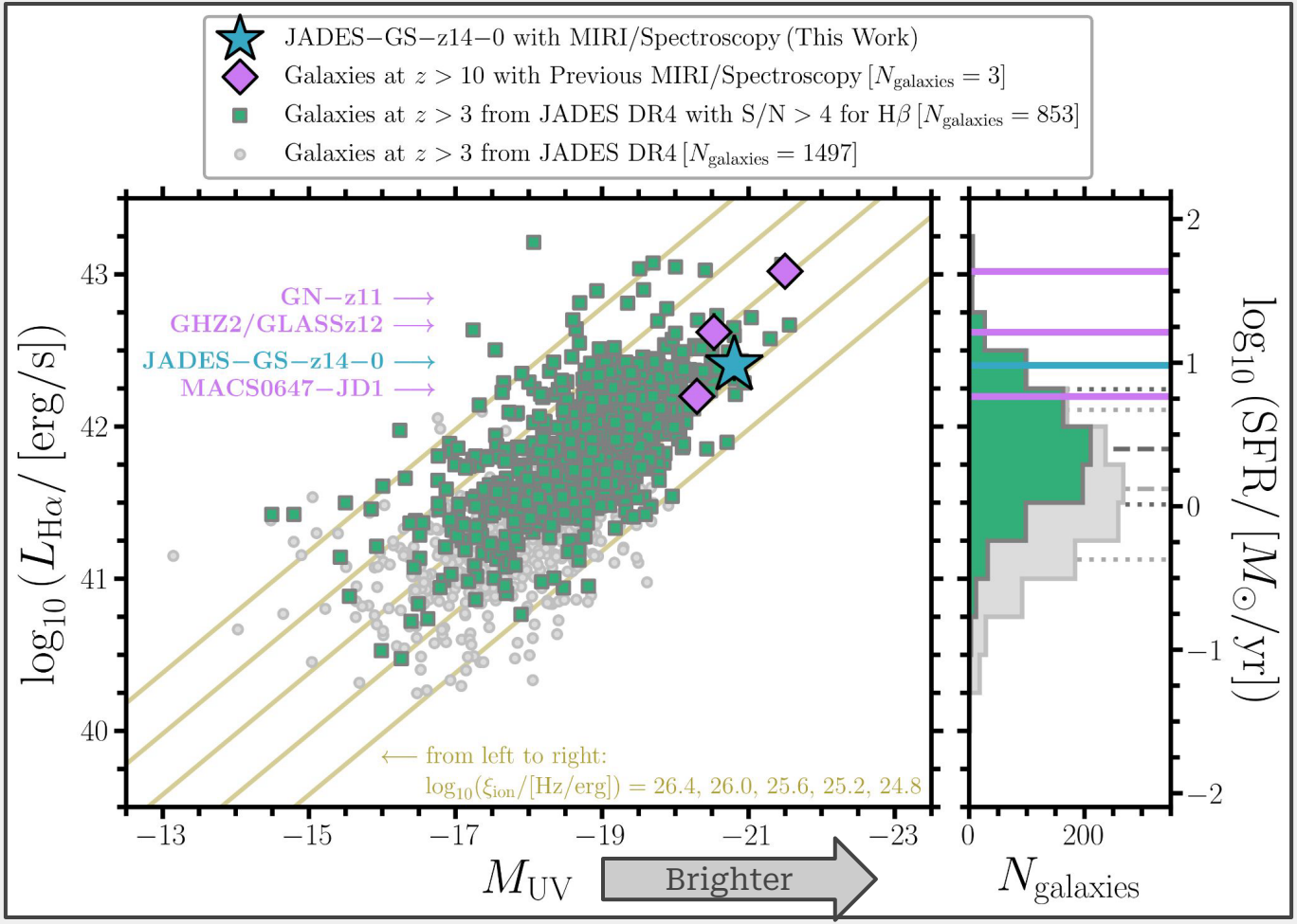


Image Credit: [Jakob M. Helton et al. (2025c)]

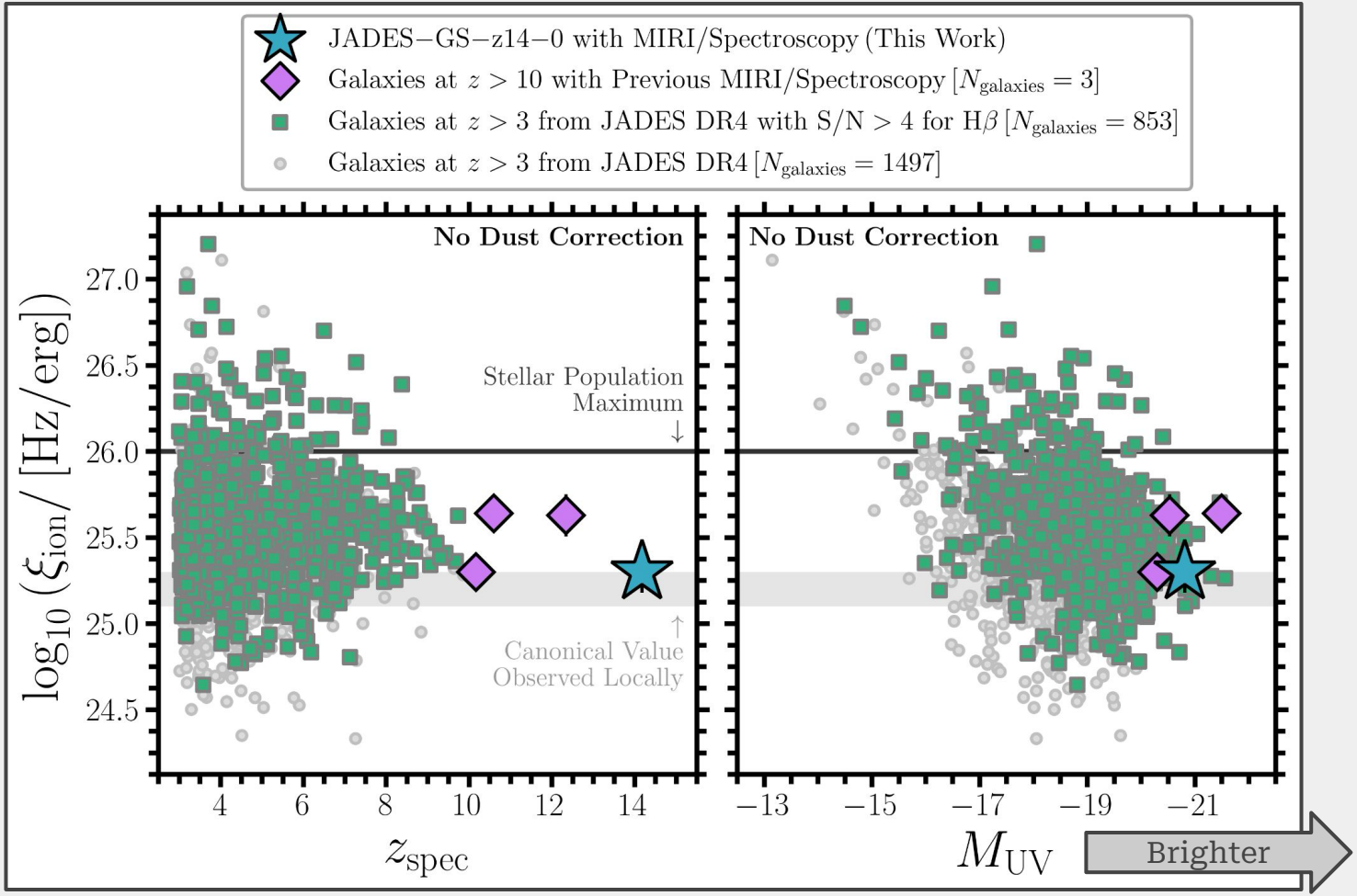


Image Credit: Jakob M. Helton et al. (2025c)

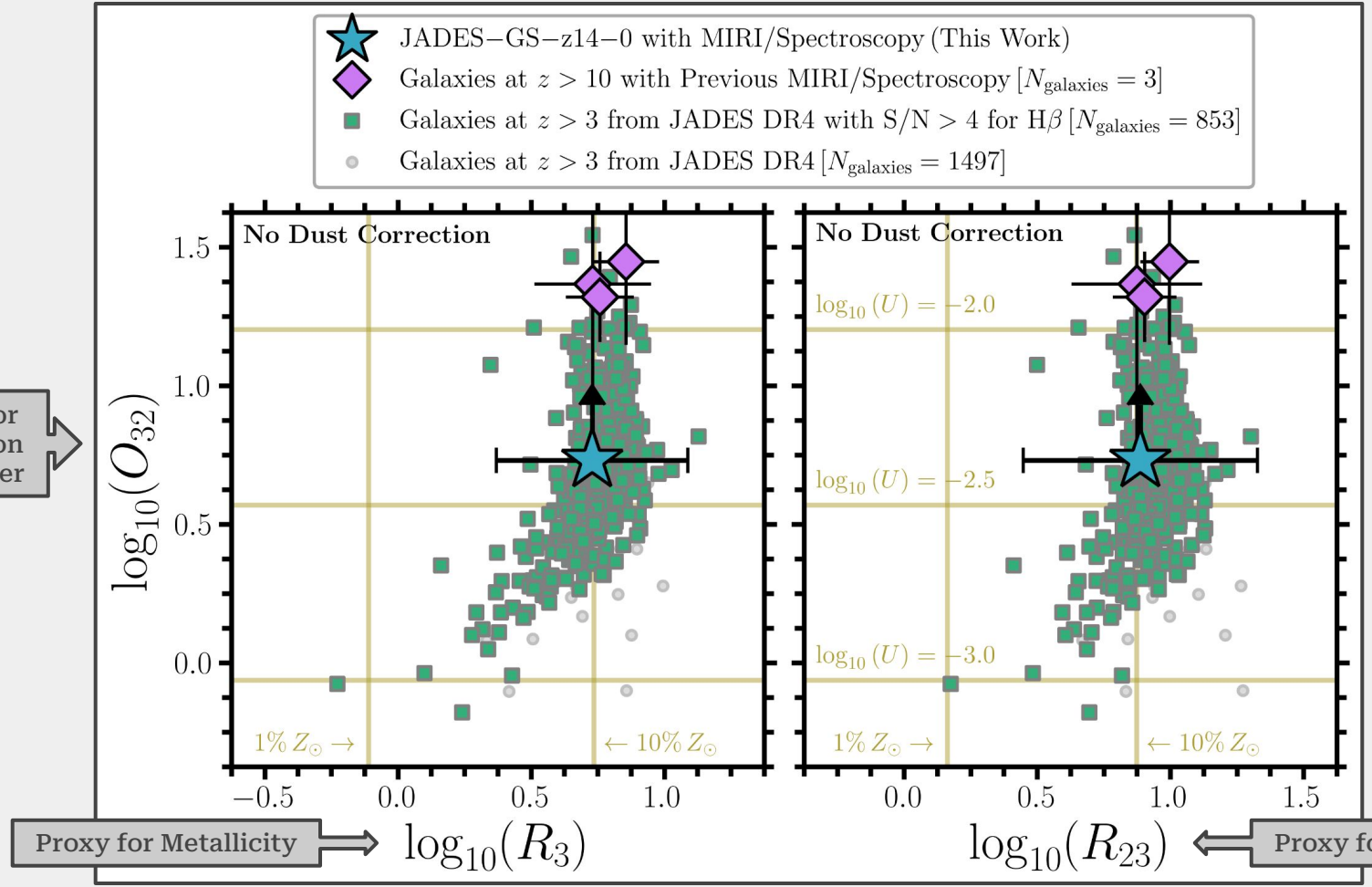
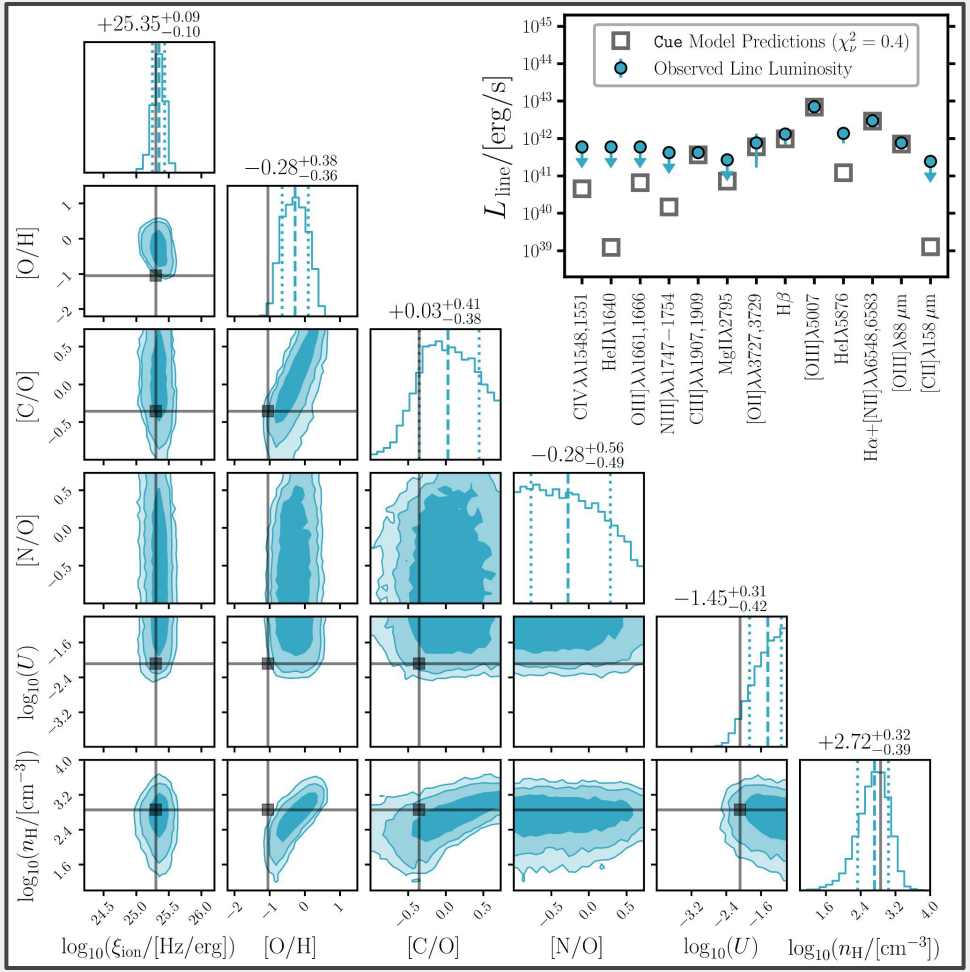


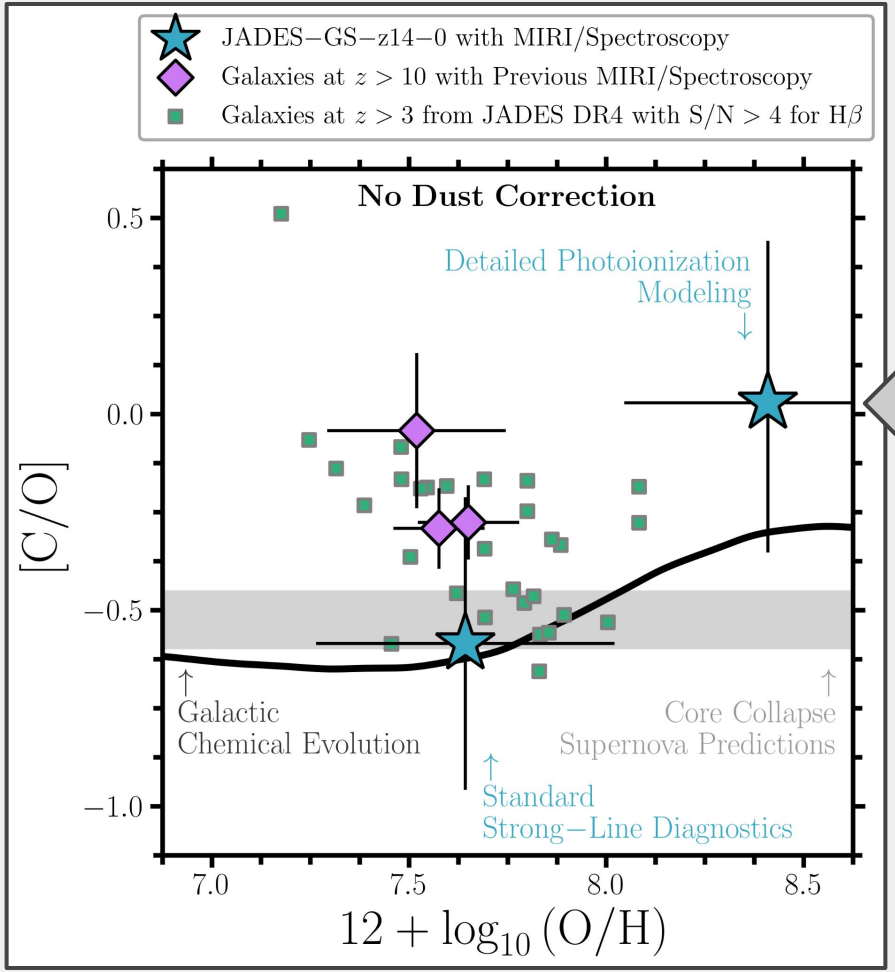
Image Credit: [Jakob M. Helton et al. (2025c)]

Detailed Photoionization Modeling Using **All** Emission Lines...



... Suggests ~50% Solar Metallicity at $z > 14$!

Image Credit: [Jakob M. Helton et al. (2025c)]



Consistent with values measured locally (stars, globular clusters, galaxies).

Image Credit: [Jakob M. Helton et al. (2025c)]

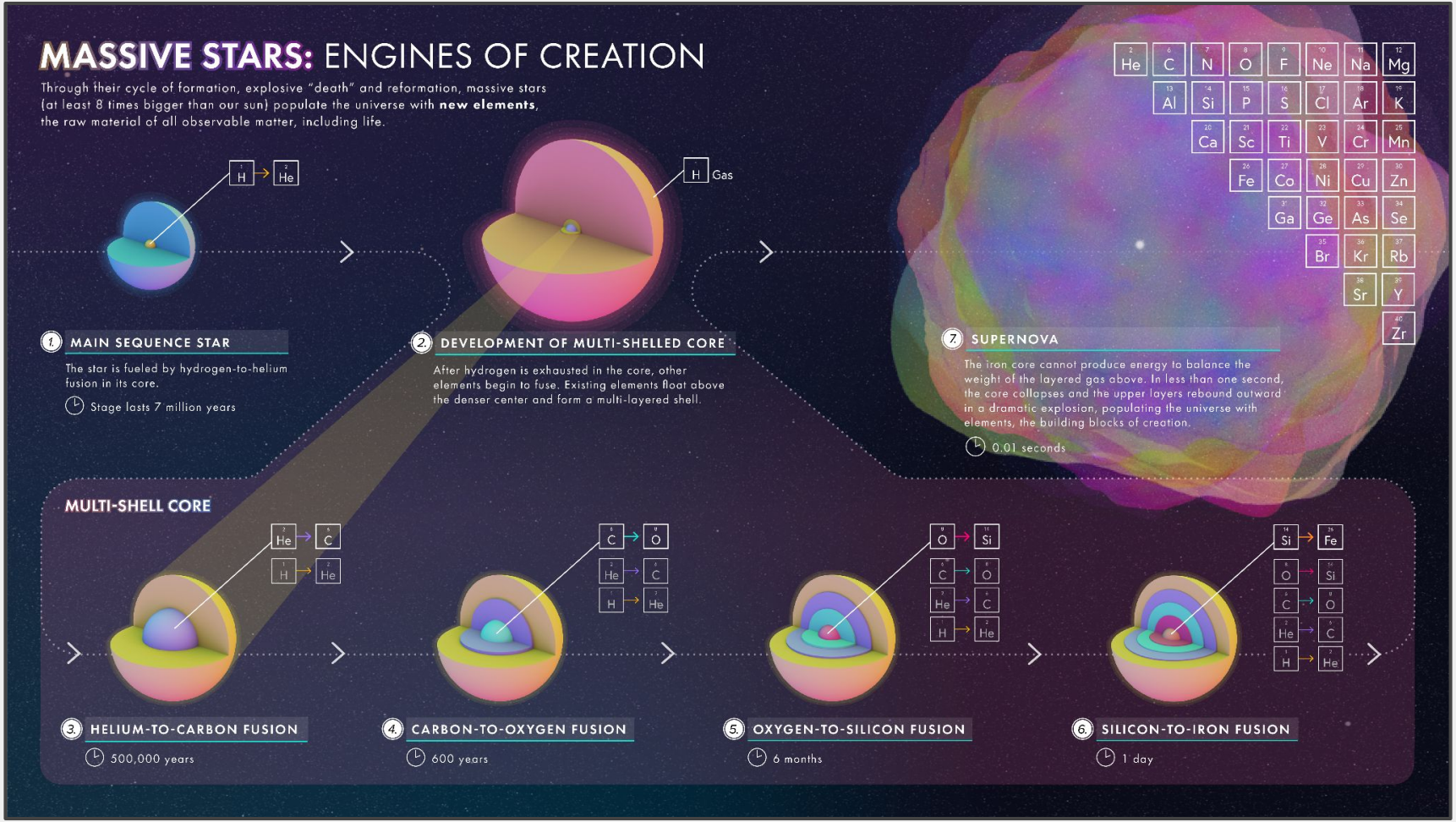
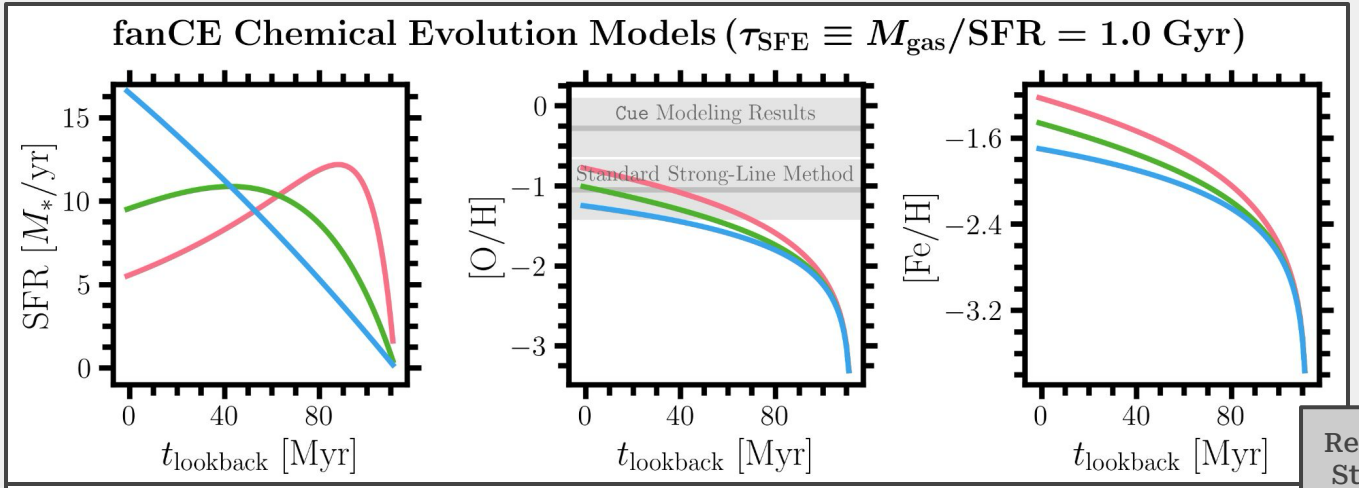
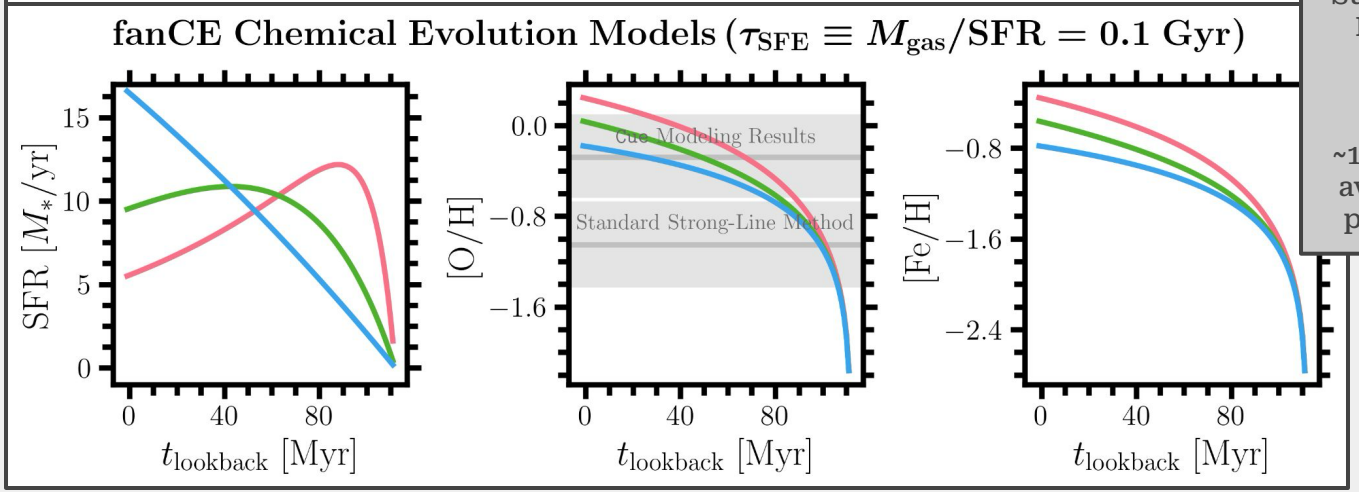


Image Credit: NASA, ESA, L. Hustak (STScI)

Star-Formation History (SFH)



Star-Formation History (SFH)

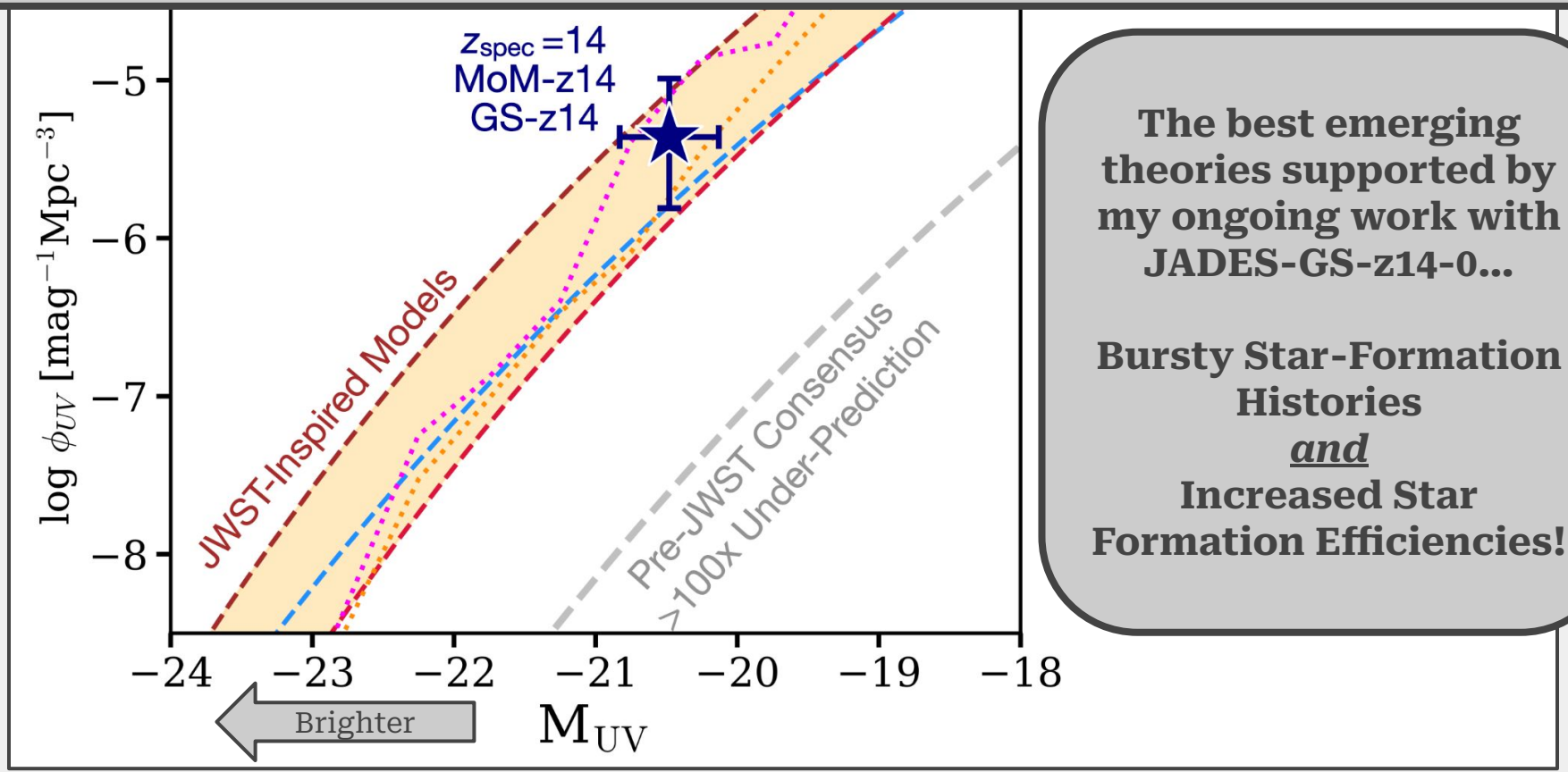


Relatively Large Star Formation Efficiencies (SFEs) are required...

~10-100% when averaged over prior 100 Myr

Image Credit: Jakob M. Helton (in preparation)

An Overabundance of Luminous Galaxies at $z > 10$? … Why?

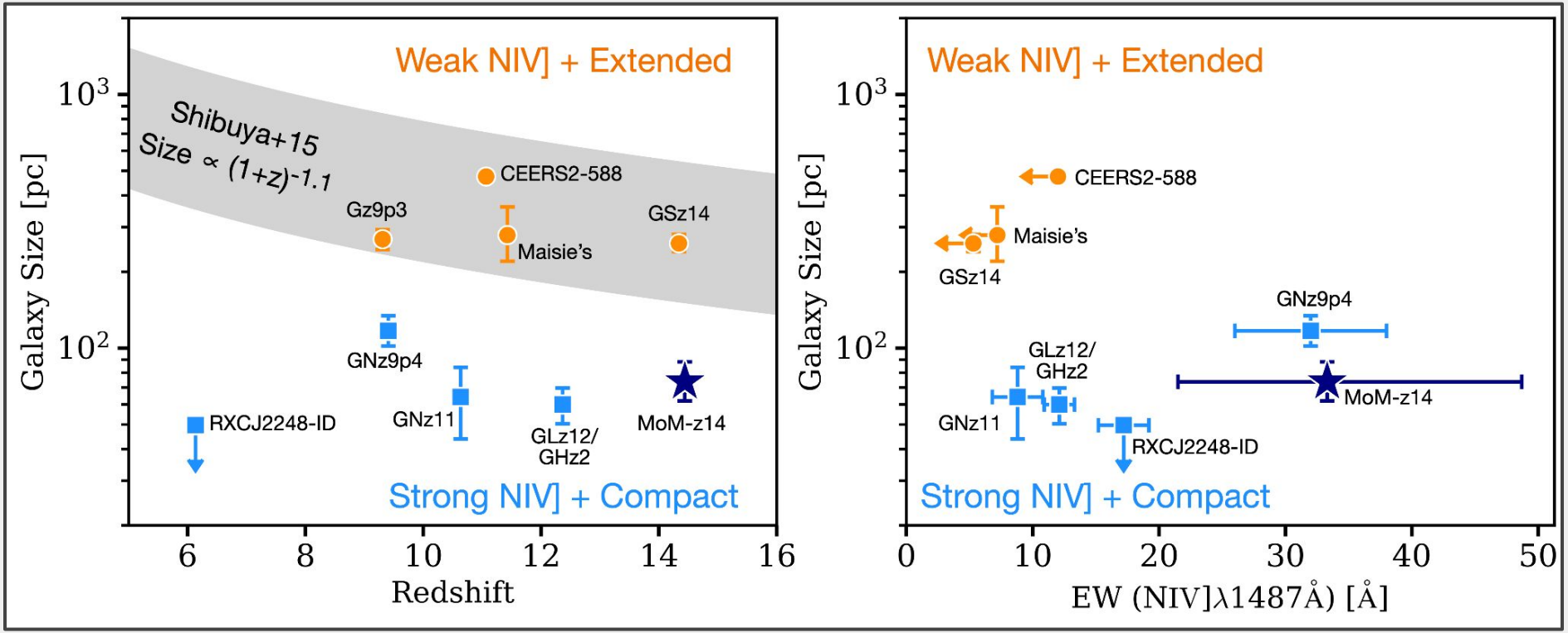


The best emerging theories supported by my ongoing work with JADES-GS-z14-0…

Bursty Star-Formation Histories *and* Increased Star Formation Efficiencies!

Image Credit: Rohan P. Naidu et al. (2026)

Two Classes of Luminous Galaxies at $z > 10$? ... Why?



Bursty Star-Formation Histories can *also* explain this puzzle!

Image Credit: Rohan P. Naidu et al. (2026)

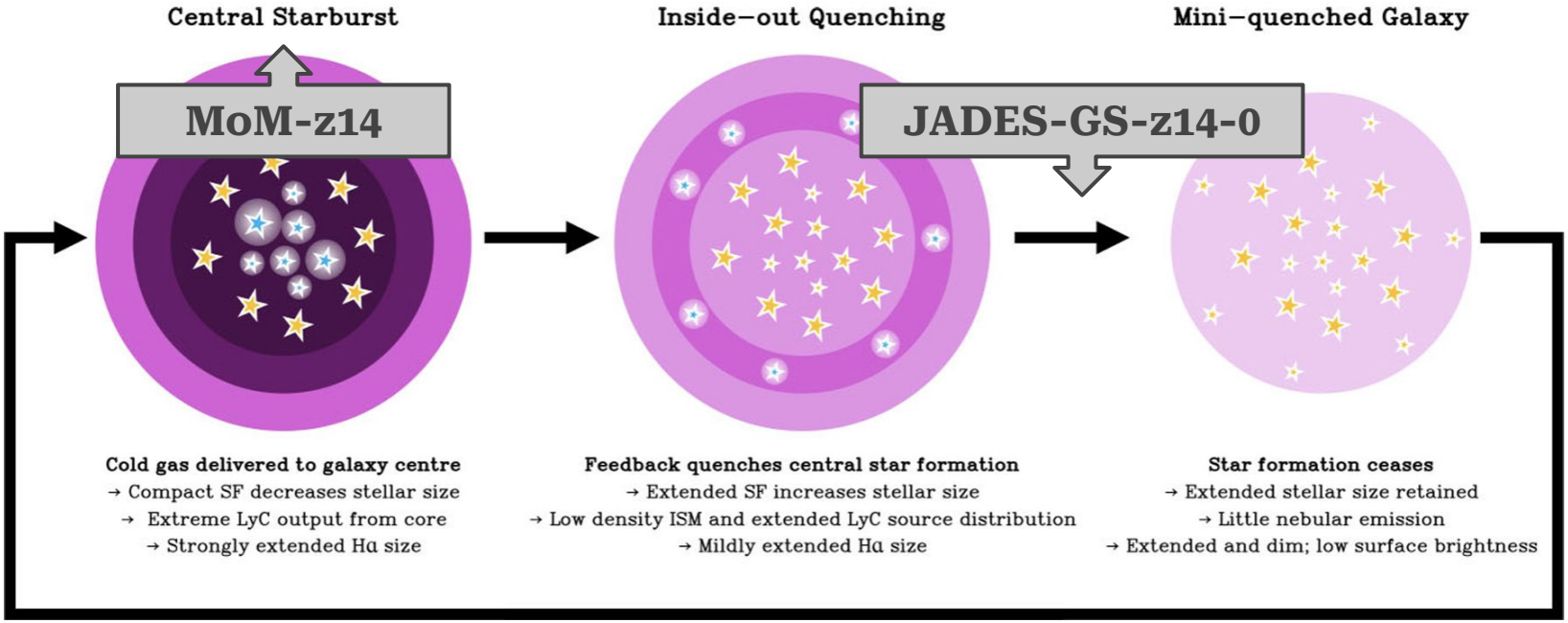


Figure 10. The evolution of galaxy sizes throughout a starburst. Blue represents young star, orange represents older stars, and the intensity of Hα emission is shown by the darkness of the purple circles. Starbursts tend to arise from the rapid supply of cold gas to the centre of a galaxy, which causes strongly concentrated star formation and the stellar size to compact drastically. The nebular emission is generally strongly extended during this period due to the extreme LyC emission from the compact star-forming core creating a large Strömgren sphere. Stellar feedback causes star formation to quench from the inside out, leading to extended star formation and an increase in the stellar size. Hα tends to be mildly extended during this phase due to a low ISM density extending the Strömgren sphere. Eventually, all star formation ceases and the galaxy retains its extended stellar size. These 'mini-quenched' galaxies show little nebular emission. They also have low continuum surface brightness due to their extended nature and dim, older stellar population.

Image Credit: William McClymont et al. (2025)

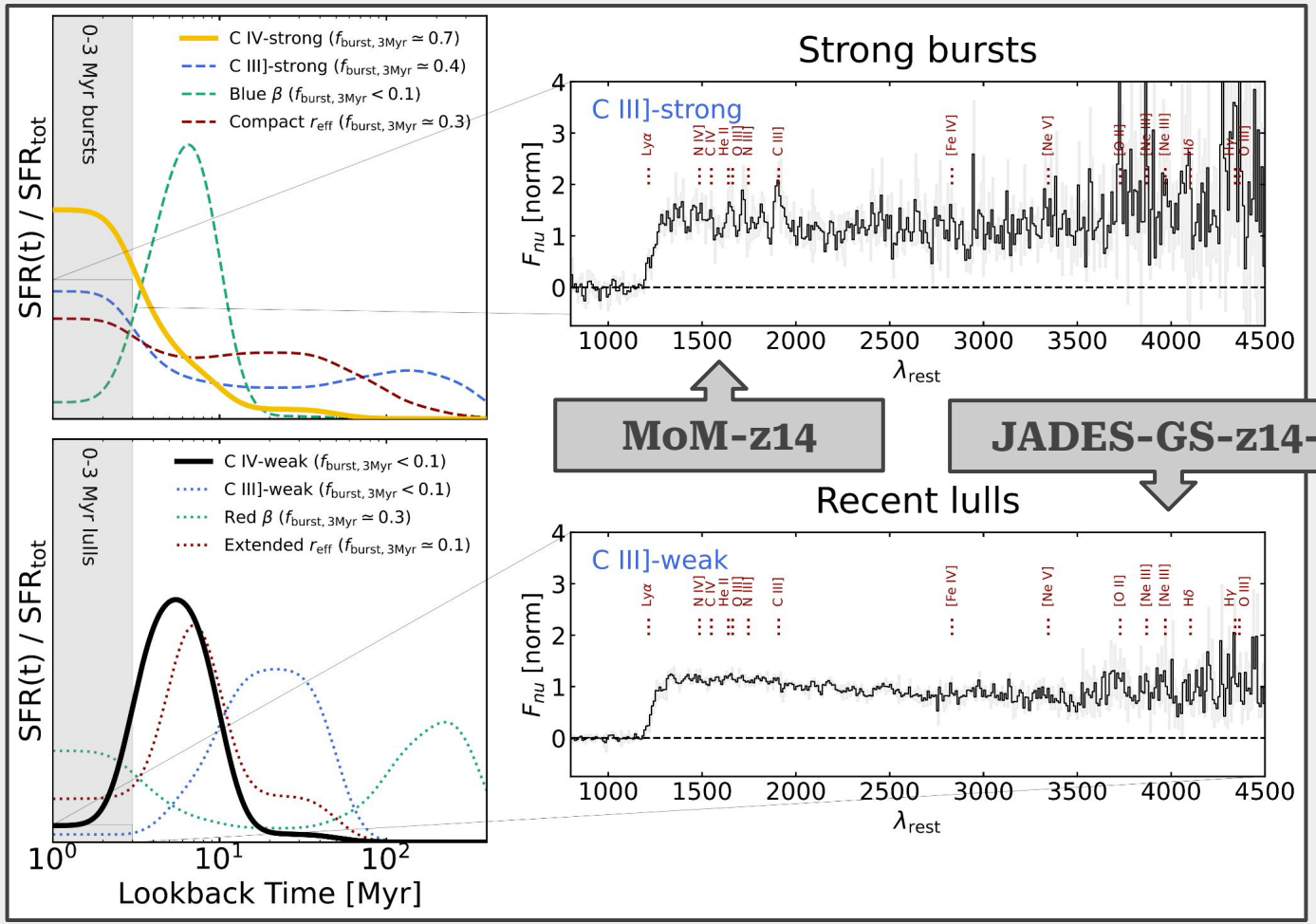


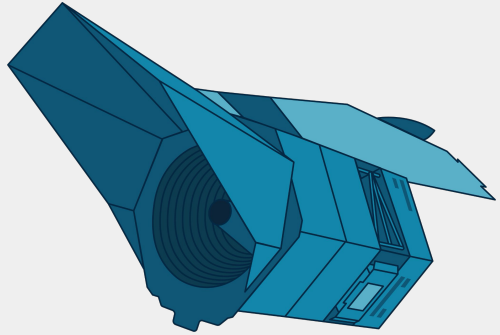
Image Credit: [*Guido Roberts-Borsani et al. (2025)*]

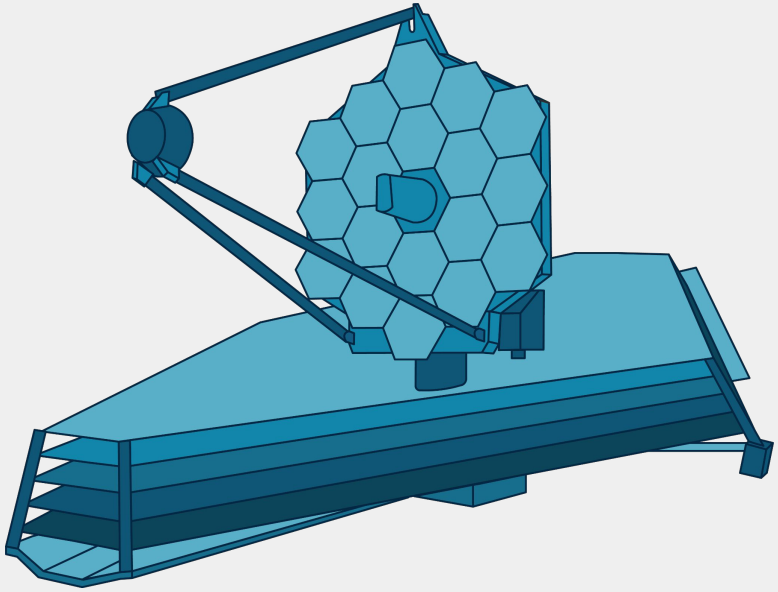
~~What are galaxies?~~

~~What is the James Webb Space Telescope (JWST)?~~

~~How is JWST being used to study the most distant galaxies?~~

Launch by May 2027





ROMAN SPACE TELESCOPE
WIDE FIELD INSTRUMENT (WFI)

WFI Field of View
2700x1380 arcseconds

WFI
Eighteen 4,096 x 4,096 pixel sensors

PRIMARY MIRROR
7.9 feet (2.4 meters)

The Nancy Grace Roman Space Telescope's primary mirror is the same size as Hubble's but only one-fourth the weight, due to advancements in technology. It will provide the same sharp resolution in infrared while surveying the cosmos hundreds of times faster.

0.5 to 2.3 micron wavelength range

0.11 arcsec/pixel
Resolution

JAMES WEBB SPACE TELESCOPE
NEAR-INFRARED CAMERA (NIRCam)

NIRCam Field of View
260x130 arcseconds

NIRCam short
Eight 2,048 x 2,048 pixel sensors

NIRCam long
Two 2,048 x 2,048 pixel sensors

0.6 to 2.3 micron wavelength range

2.4 to 5.0 micron wavelength range

0.031 arcsec/pixel
Resolution

0.063 arcsec/pixel
Resolution

PRIMARY MIRROR
21 feet (6.5 meters)

Webb's mirror consists of 18 panels made of beryllium and thinly coated with gold. Webb's mirror will have more than six times the light collection of Hubble and Roman.

SHORTEST WAVELENGTH: 0.5 MICRON (BLUE-GREE
LONGEST WAVELENGTH: 2.3 MICRONS (NEAR-INFRA
SHORTEST WAVELENGTH: 0.6 MICRON (YELLOW)
LONGEST WAVELENGTH: 5.0 MICRONS (MID-INFRARED)

Image Credit: [NASA's Goddard Space Flight Center]

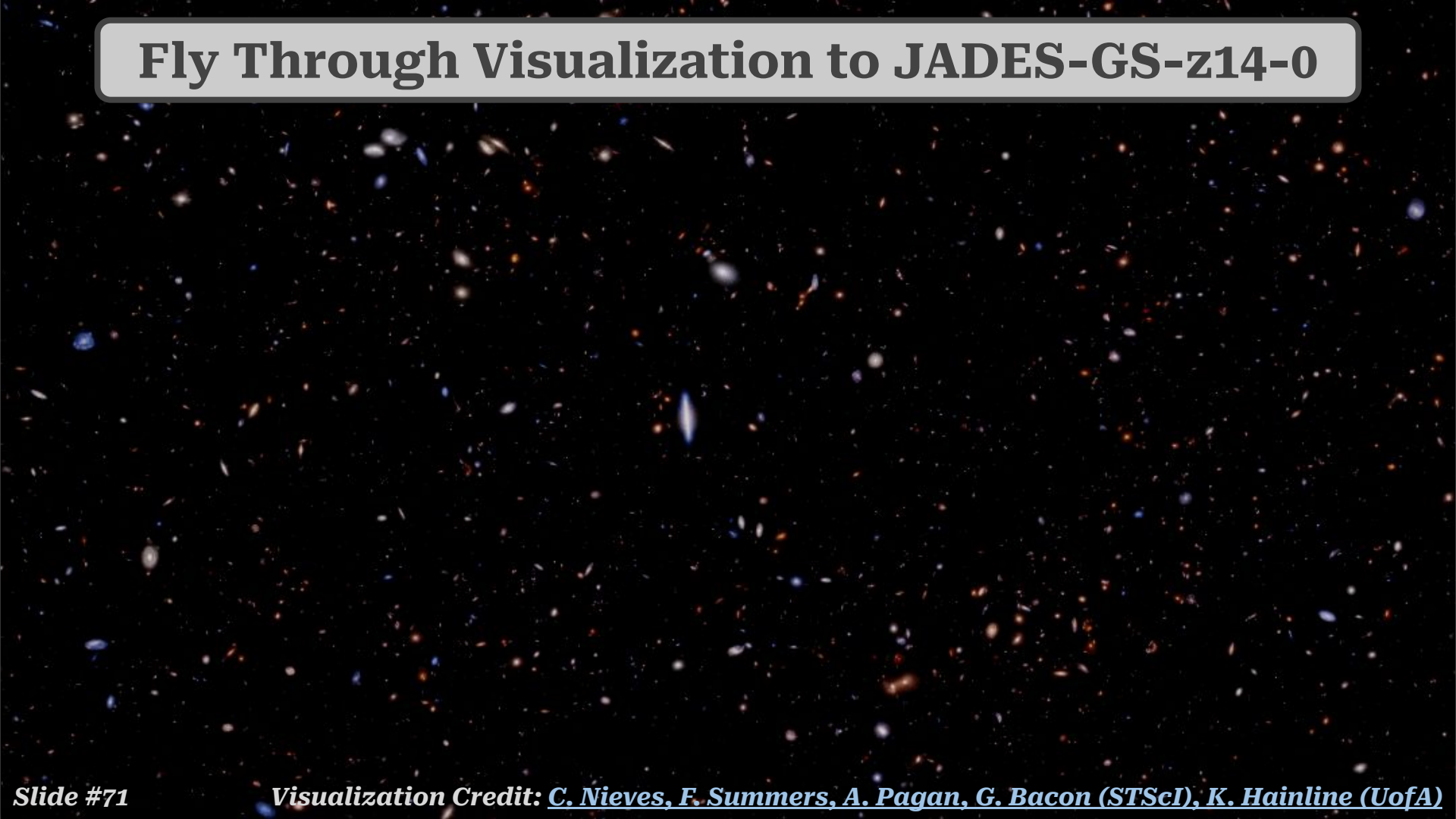
– Summary and Central Thesis –

- The most distant galaxy candidates are found with **JWST/NIRCam** (and eventually will be found with **Roman**).
- Then they are confirmed with **JWST/NIRSpec** or **ALMA**.
- But they are not fully understood without **JWST/MIRI**.

JWST/MIRI is still the future for understanding the first stars and galaxies!

Image Credit: NASA, ESA, CSA, the JADES Collaboration, A. Pagan (STScI)

Fly Through Visualization to JADES-GS-z14-0



Visualization Credit: C. Nieves, F. Summers, A. Pagan, G. Bacon (STScI), K. Hainline (UofA)

At the Break of Cosmic Dawn: Identifying and Understanding the Most Distant Galaxies with JWST

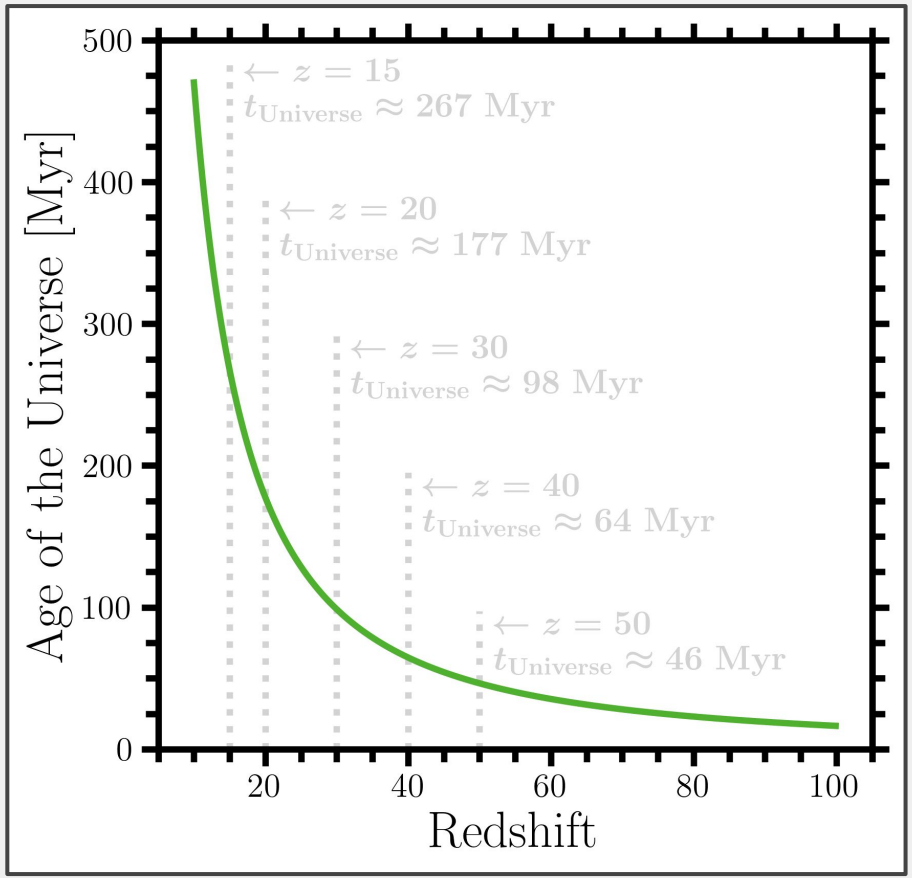
Dr. Jakob M. Helton (jakobhelton@psu.edu)

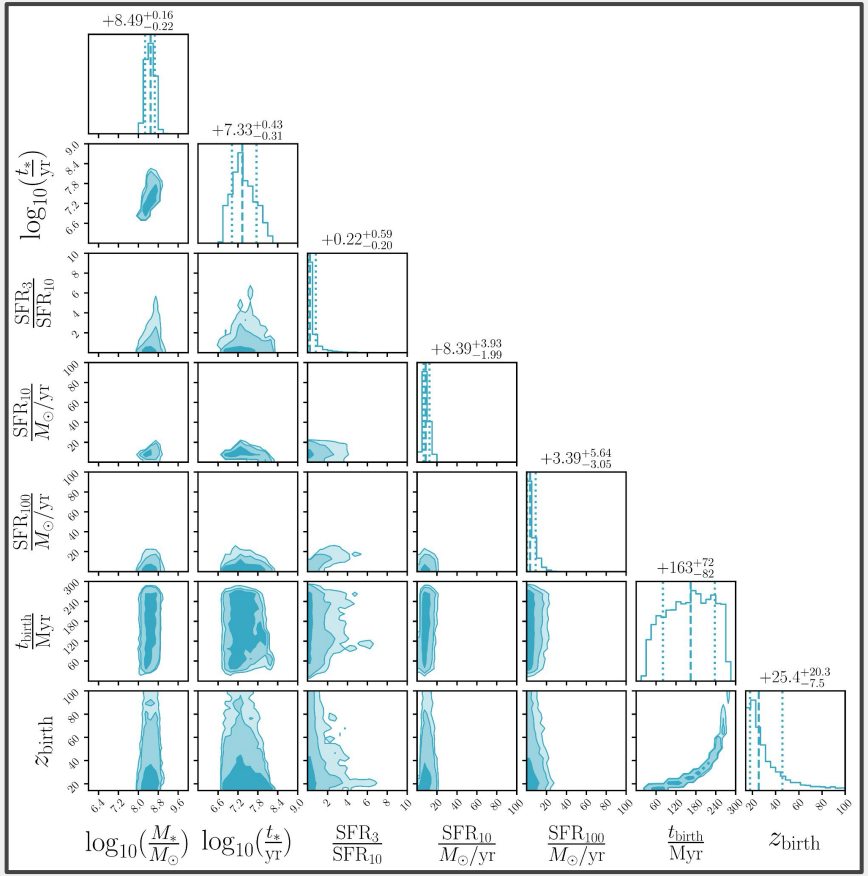
Evolving Universe Postdoctoral Fellow

PennState Eberly College of Science

JADES JWST ADVANCED DEEP EXTRAGALACTIC SURVEY

Can we constrain when the *first stars* formed? ... Maybe!





Backup

Image Credit: Jakob M. Helton (in preparation)